

Universal Insurance with In-Kind Transfers: The Welfare Effects of Long-Term Care Insurance in Japan *

Minamo Mikoshiba [†]

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Abstract

This study quantifies the welfare effects of public long-term care insurance (LTCI) system in Japan, focusing on the role of the universal insurance system with benefit-in-kind, in a rich overlapping generations model with two-sided altruism. The universal LTCI with in-kind benefits is more expensive than a cash benefits system although the in-kind policy positively impacts caregivers' labor supply. The welfare gains from cash benefits depend on caregiver productivity; when productivity is low, cash benefits can maintain positive welfare effects while reducing government LTCI expenditure. The elimination of universal LTCI shifts the burden of caregiving to families and increases the reliance on means-tested welfare programs, which distorts savings behavior and offsets the reduction in government spending.

Keywords: social security, long-term care, long-term care insurance, overlapping generations model, Japan.

JEL Classification: D15, E6, I10, I13, J14

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[†]Nagoya University, Japan. Email address: minamo.mikoshiba@gmail.com

1 Introduction

Developed countries have been experiencing population aging, with Japan at the forefront of unprecedented demographic aging. This ongoing demographic aging has resulted in a higher risk of disability. For example, older adults have difficulty performing activities of daily living (ADL) and instrumental activities of daily living (IADL) (Christensen et al. 2009; Chatterji et al. 2015). In Japan, no less than 18% of individuals aged 65 years and above, and at least 60% of those above 85 years of age, are officially certified as needing long-term care or support, according to the Report Survey on Situation of Long-term Care Insurance Service of the Ministry of Health, Labour and Welfare (MHLW) in 2019. With the increasing proportion of older adults in the late stage, the Japanese government is forecasting an upsurge in the burden imposed on families and the expenses related to social security programs. This growing number of older adults has put immense pressure on the government to find ways to deal with their caregiving needs.

In this context, this study establishes the following two objectives. Firstly, this study aims to analyze the impact of the risk of long-term care on individuals' behavior over the life-cycle. Once older adults need long-term care, they and their family members face significant care burdens through time and/or long-term care expenditures. In other words, older adults with disabilities and their families face a trade-off between unpaid care by family members (informal care, hereinafter, IC) and long-term care services that are provided by the market (formal care), such as at-home care (formal home care, hereinafter, FHC)¹ and nursing home care (NHC).² The informal care imposes a significant time burden on the family caregivers and disrupts the labor supply of family caregivers, especially if they belong to working-age. The formal care imposes a significant financial burden on the older adults with disabilities and their families. This study analyzes care arrangements focusing on three care options: IC, FHC, and public NHC. As IC and FHC are available at home, this study considers the substitutability of the two when considering care arrangements. The second objective of this study is to quantify the welfare cost of Japan's LTCI system in comparison to alternative policies, to evaluate the role of the system. This study focuses on the unique characteristics of Japan's LTCI system, which is a universal coverage insurance and benefits-in-kind system. Japan's LTCI system covers all citizens and provides them with an identical set of benefits-in-kind. Unlike other

¹As documented in Fu et al. (2017), FHC services include housekeeping, bathing, visiting nurses, rehabilitation, day services, short-stay services, medical care management counseling, welfare device leasing/purchasing, and home renovation. In contrast, nursing home care, as well as chronic care hospitals, is included in institutional services.

²This study does not consider private NHC services as long-term care options. This is because public NHC accounts for most of the total facility capacity, and information on occupancy rates in private facilities is not available in sufficient detail. Website: https://www.mhlw.go.jp/file/05-Shingikai-12601000-Seisakutoukatsukan-Sanjikanshitsu_Shakaihoshoutantou/0000171814.pdf (in Japanese) (Accessed July 10, 2022).

countries with social insurance for long-term care policies, such as Germany and South Korea, Japan's LTCI system only provides services and no cash allowances.

To analyze how disability risks affect individuals' life-cycle behavior, this study incorporates distinct trajectory patterns of disability and mortality risks into the structural model. This study uses one-year interval transition probabilities, estimated by [Mikoshiha et al. \(2023\)](#) from the nationwide administrative claims data of LTCI system in Japan. To the best of my knowledge, this study is the first study to use the LTCI administrative claims data in a rich structural model. Since the eligibility for LTCI is determined solely based on objectively measured demands for care, this study has substantial advantages in terms of the accurate assessment of the risks over self-reported disability status. Since the administrative claims data does not include sociodemographic and socioeconomic information, this study uses the Comprehensive Survey of Living Conditions (CSLC) to capture patterns of care arrangement among older adults with disabilities.

By using estimated transition probabilities, this study can capture the heterogeneity of disability risks in old age and the different trajectories of transitions. Although disability risks are sometimes studied and discussed together with medical expenditure risks, what should be emphasized is the almost irreversible persistence of the risks. Once individuals are in need of care, it is difficult to return to an independent daily life. Another different aspect from the medical expenditures risks is the option of home production for care.

Using the CSLC, this study finds that approximately 80% of the LTCI recipients receive care services at home. Although about 70% of LTCI recipients at home use a mix of IC and FHC, family members are the key caregiving source. IC accounts for most primary caregivers, especially children, and the burdens of caregivers are concentrated on them. Long-term care problems are similar to child care problems in terms of the importance of family care, but care for older adults differs in terms of the directions of the care from the adult children to their older parents.

To capture these empirical facts, this study develops a rich structural overlapping generations model with two-sided altruism. This model is populated by heterogeneous families differing in various dimensions. The sources of uncertainty in this model are disability and longevity risk in old age, permanent skill shocks, and idiosyncratic wage shocks during working-age. There is no insurance market for these risks, and families face a no-borrowing constraint. The adult child and older parent generation from families jointly make decisions to maximize the same objective function in the sense of two-sided altruism (e.g., [Fuster et al. \(2007\)](#) and [İmrohoroglu and Zhao \(2018\)](#)). The family chooses optimal allocations for life-cycle consumption, caregiver labor supply, savings, and care arrangements. My model focuses on the disability status of the female members in the older parent generation and the care arrangements between the female older parent and the female adult child as her primary caregiver. To develop a richer model of care arrangements, I incorporate three types of care options: IC from her female adult child, FHC,

and public NHC. Furthermore, this model endogenizes care arrangements by introducing two-stage family decisions.

This study calibrates the model parameters to the Japanese economy in 2015. Relative to the pattern of care arrangements in the data, the model replicates the overall pattern of care arrangements well. This study evaluates the universal insurance system with benefits-in-kind by quantifying the welfare effects of the LTCI relative to alternative long-term care policies. First, this study considers the universal insurance system with cash benefits. This study sets the amount of cash benefits to be equal to the government expenditures on LTCI for each level of care needs in the baseline model. This study also considers the elimination of the LTCI system from the economy to evaluate the role of the universal insurance system. Since long-term care expenditures for welfare transfer recipients are financed through means-tested welfare programs, the mean-tested welfare program becomes the only long-term care policy in an economy without LTCI.

For the policy experiments, I find that universal LTCI with in-kind benefits is more expensive than a cash benefit system, although the in-kind policy does not discourage caregivers' labor supply. Under an alternative scenario in which LTCI provides only cash benefits with a 100% copayment rate, the cost of formal care, such as FHC and NHC services, becomes higher and caregivers' labor supply to provide IC. This reduces labor income tax revenues, but the impact of tax adjustment is limited for low productivity of caregivers. This positive welfare gain from cash benefits holds when the government reduces its spending on LTCI by 10%. In contrast, eliminating universal LTCI shifts the burden of caregiving to families and increases reliance on means-tested welfare programs. The sign of the welfare effects depends on the level of the consumption floor through tax adjustments. When the level of the consumption floor is small, the welfare loss from the distortion of saving is dominated by welfare gains for the sufficient compensation through tax adjustment effects.

This study builds on multiple lines of literature. This study is related to a huge body of literature investigating the roles of government insurance policies for older adults. Starting with [Auerbach and Kotlikoff \(1987\)](#), a number of literatures have analyzed the roles of social security systems.³ In particular, this study contributes to the growing literature investigating the roles of long-term care policies. [Barczyk and Kredler \(2017\)](#) focus on means-tested Medicaid and evaluate non-means-tested IC and formal care subsidies as alternative scenarios. They show that the combination of IC and formal care subsidies precipitates a large welfare gain, reduction in fiscal spending on Medicaid, and decline in labor income tax revenues. This is because low-income individuals respond to IC subsidies,

³Recent studies have focused on the roles of insurance policies in health and medical expenditure risks. Among the earlier studies, [Attanasio et al. \(2010\)](#) consider the role of Medicare, [Braun et al. \(2017\)](#) examine the optimal size of means-tested Medicaid, and [Pashchenko \(2013\)](#) evaluate the implications of the Affordable Care Act.

and a decline in their labor income tax revenues modestly impacts the total tax revenues. In other streams of literature, some studies analyze the interaction between Medicaid and the long-term care market. [Brown and Finkelstein \(2008\)](#), [Mommaerts \(2025\)](#), [Ko \(2022\)](#), and [Braun et al. \(2019\)](#) discuss the demand for Medicaid and the private LTCI market in the U.S. [Koreshkova and Lee \(2020\)](#) study the interactions of Medicaid and the nursing home care market. These studies on means-tested transfer help us understand the role of means-tested transfer as a long-term care policy in macroeconomics. However, most developed countries do not have means-tested transfers and have a universal LTCI system with a benefits-in-kind policy or/and cash benefits for the long-term care policy, as shown in [Barczyk and Kredler \(2019\)](#). This study contributes to the literature by evaluating long-term care policies, such as universal LTCI systems with a benefits-in-kind policy, by considering means-tested welfare programs and cash transfers as alternative policies. Furthermore, this study contributes to the literature by incorporating three types of care options, that is, IC from her adult child, FHC, and public NHC, and by endogenizing care arrangements by introducing two-stage family decisions. Previous studies using U.S. data usually focus only on IC and NHC and dismiss FHC. However, it is important to consider FHC when analyzing the role of long-term care policies because NHC is more expensive than FHC for the government.

Second, this study also contributes to the literature on the effect of long-term care risk on households' life-cycle behavior. In macroeconomics literature, precautionary savings against long-term care risks in old age are widely discussed. Previous studies by [Kopecky and Koreshkova \(2014\)](#), [Ameriks et al. \(2020\)](#), and [Bueren \(2022\)](#) show that the impending risk of future long-term care spending induces older parents to hold onto assets for self-insurance. [De Nardi et al. \(2010\)](#) find that spending on health care later in life is important in explaining the slow decline in spending during retirement. [Lockwood \(2018\)](#) shows that precautionary savings against long-term care risk lead to large bequests when individuals do not need long-term care. [İmrohoroglu and Zhao \(2018\)](#) introduce a dynastic framework and show that households cope with long-term care risk by increasing savings when the family insurance channel weakens.⁴

This study also relates to the empirical applied microeconomic literature regarding care arrangements. Various studies have revealed that IC negatively impacts caregivers' labor force participation. [Van Houtven et al. \(2013\)](#) and [Skira \(2015\)](#) find that caregivers in the U.S. are less likely to work than non-caregivers. Furthermore, [Sugawara and Nakamura \(2014\)](#) reveal the negative impact using Japanese data.⁵ [Fu et al. \(2017\)](#) reveal that Japan's LTCI system with benefits-in-kind exerts significant and positive spillover effects

⁴Several studies have analyzed housing and LTCI. [Davidoff \(2010\)](#) study the substitution relation between home equity and LTCI. [Barczyk et al. \(2022\)](#) examine the interactions between housing and the family for the saving and intergenerational transfer behavior of the elderly.

⁵See also [Shimizutani et al. \(2008\)](#), [Hanaoka and Norton \(2008\)](#), [Yamada and Shimizutani \(2015\)](#), and [Ando et al. \(2021\)](#) for more studies in Japan.

on caregivers' labor force participation. In contrast, using German data, [Geyer and Korfhage \(2015\)](#) demonstrate that the LTCI system with cash transfers negatively impacts labor force participation. Moreover, several studies in the U.S. and European countries report that IC is substituted for formal care. [Charles and Sevak \(2005\)](#) provide strong evidence of substitution between IC and NHC. [Bonsang \(2009\)](#) reveal that IC and FHC are partial substitutes.⁶ Although the FHC policy would mitigate the burden on family caregivers, careful consideration must be given to the substitutability or complementarity between IC and FHC to evaluate the policy.

Third, recent studies have built life-cycle models including medical and long-term care expenditures to analyze fiscal sustainability in Japan under demographic aging. Some studies have used publicly available data and a deterministic process of expenditures,⁷ while few studies have incorporated stochastic health risks using the administrative claims data. [Hsu and Yamada \(2019\)](#), [Fukai et al. \(2021\)](#) and [Hagiwara \(2022\)](#) use claims data to analyze the welfare effects of health insurance reform. The analysis in this study parallels that of [Fukai et al. \(2021\)](#). However, it is worth noting that while medical treatment is performed in hospitals and medical facilities by qualified individuals, individuals with disabilities often receive informal care at home from family members in addition to formal care. This study focuses on incorporating both informal and formal care into the model and endogenizing care arrangements between female older parents and their adult daughters based on the current disability status. The study also endogenizes the labor supply of family caregivers, which is an important factor to consider when analyzing the impact of long-term care risk on households' life-cycle behavior.

The remainder of this study is organized as follows: Section 2 describes Japan's LTCI system; Section 3 documents the empirical findings on long-term care risks over the life-cycle and care arrangements; Section 4 presents the quantitative life-cycle model; Section 5 describes the model's parametrization; Section 6 presents the numerical results; Section 7 presents the concluding remarks.

2 Public Long-Term Care Insurance System in Japan

Japan implemented the LTCI in 2000. Japan then became one of the first countries to develop mandatory public insurance schemes for long-term care, both at home and in nursing homes.⁸ Japan's LTCI system was introduced to help older adults with disabilities

⁶See also [Van Houtven and Norton \(2004\)](#), [Bolin et al. \(2008\)](#), and [Mommaerts \(2018\)](#) for more studies in this regard.

⁷See, for example, [Braun and Joines \(2015\)](#), [Kitao \(2015\)](#), and [İmrohoroglu et al. \(2016\)](#).

⁸For details on Japan's LTCI system, including its history, see [Campbell and Ikegami \(2000\)](#) and [Tamiya et al. \(2011\)](#). For further details of the program, see the description on the MHLW website: <https://www.mhlw.go.jp/english/topics/elderly/care/index.html> (As of July 2002) (Accessed September 13, 2024).

“to maintain dignity and an independent daily life routine according to each individual’s own abilities,” according to the Long-term Care Insurance Act (*Kaigo hoken hō*) from the Ministry of Justice.

The insured persons consist of those who are 65 years and above and those who are 40 to 64 years of age who are insured by the health care insurance system. This study focuses on individuals aged 65 and above, accounting for approximately 98% of total recipients (about 6.82 million). The insured persons under the LTCI system do not necessarily coincide with the recipients.

It is important that the LTCI eligibility is determined solely based on the level of care demanded, regardless of socioeconomic attributes, such as family and economic status, without means tests.⁹ The LTCI quantifies the level of care demanded by calculating the standard hours of total care demanded. Standard hours of care are estimated based on 74-item questionnaires on ADL, IADL, behavioral and psychological symptoms of dementia (BPSD), functional training, and medical care. Further, based on standard hours of care, each older adult is classified into one of the eight levels of care needs by a computer algorithm and an expert committee. Once classified as eligible for LTCI, reassessments are conducted every year in principle.¹⁰

The eight levels of care demanded consist of ineligible or independent, support-required level (SL)1-2, care-required level (CL)1-5. If classified as ineligible or independent, the older adults are not eligible for LTCI. SL1 is the mildest and CL5 is the severest level of care needs. SL refers to the recipients who live independently but need help with IADL. Care recipients in SL1-2 are eligible for preventive care services to keep recipients from reaching a more severe care levels. CL refers to the recipients who require greater help with ADL and IADL. CL1 and CL2 include those who might be able to live alone if provided partial assistance with basic activities, but who are observed with lower levels of thinking and some problematic behavior. In contrast, CL3-5 are assigned to more severe older adults who are unable to live without full support for daily life. Table 1 shows the activities and conditions that about 80% of eligible individuals in each level of care require some assistance.

⁹Welfare transfer program recipients who are above the age of 65 are eligible to use LTCI services if they are certified as needing long-term care or support. Their LTCI premiums are financed by livelihood assistance. Out-of-pocket long-term care service expenditures for them are covered by long-term care assistance.

¹⁰For those who are certified for the first time, reassessments are performed six months following the first certification. Individuals can request a reassessment if they experience a decline in health or have questions about the assessment’s results. See [Tsutsui and Muramatsu \(2005\)](#) for further details of the certification process.

Table 1: Approximate Conditions by Levels of Care Demanded

Activity/condition requiring some assistance	SL1	SL2	CL1	CL2	CL3	CL4	CL5
Getting up on their own	✓	✓	✓	✓	✓	✓	✓
Standing up on their own	✓	✓	✓	✓	✓	✓	✓
Standing on one leg		✓	✓	✓	✓	✓	✓
Shopping		✓	✓	✓	✓	✓	✓
Making Decision/Comprehension/Thinking		✓	✓	✓	✓	✓	✓
Walking				✓	✓	✓	✓
Bathing/Showering				✓	✓	✓	✓
Clipping their own fingernails				✓	✓	✓	✓
Managing medications				✓	✓	✓	✓
Managing finances				✓	✓	✓	✓
Preparing simple meals				✓	✓	✓	✓
Turning over while sleeping					✓	✓	✓
Controlling their bladder and bowel function					✓	✓	✓
Oral hygiene					✓	✓	✓
Dressing/Undressing					✓	✓	✓
Holding a sitting position						✓	✓
Standing on both legs						✓	✓
Transferring/Moving around						✓	✓
Facial cleansing/Hairdressing						✓	✓
Eating meals							✓
Short-term memory							✓
Paralysis (left lower extremity)							✓

Source: MHLW. Website: <https://www.mhlw.go.jp/file/05-Shingikai-11901000-Koyoukintoujidoukateikyoku-Soumuka/0000126240.pdf> (in Japanese, page 10 and 11) (Accessed September 13, 2024).

Note: Table 1 shows the tabulation of items from questionnaires in which more than 80% of respondents selected some assistance in each level of care, not a definition of conditions based on the level of care.

Japan's LTCI aims to relieve family caregivers' burden and emphasize at-home care rather than nursing homes. A notable feature of Japan's LTCI is its focus on providing care services, without cash allowances. Although other countries, such as Germany and South Korea, provide both services and cash allowances, Japan provides only services. This comes from concerns that family caregivers would continue to be exploited if cash allowances are given.¹¹ Japan's LTCI system sets the maximum amount of services that can be purchased as benefits for each level of care. Within the ceiling amount, the LTCI covers 90% of expenditures and sets the copayment rate at 10%.¹² Eligible individuals can

¹¹According to Campbell and Ikegami (2000) and Tamiya et al. (2011).

¹²The ratio exceeding the ceiling number of benefits is extremely low, 1.3% of all recipients in 2015,

choose the type of care services from the long-term care market, such as FHC and NHC. The Japan’s LTCI is then designed to emphasize at-home care over nursing homes, driven by the growing demand for care and substantial fiscal concerns. The primary public NHC, i.e., welfare care facilities for the elderly (*tokubetsu-yogo-rojin-home*), are only available to older adults with CL3-5—except when the level of care improves after entering nursing homes or family circumstances are serious. The LTCI does not cover living expenses and meal costs in nursing homes.

3 Long-Term Care Risks and Care Arrangements

3.1 Data Description

To gain insights into the disability and mortality risk profiles in old age, as well as the care arrangement patterns among older adults with disabilities, this study uses two primary data sources: the Statistics of Long-term Care Benefit Expenditures (SLBE) and the Comprehensive Survey of Living Conditions (CSLC), both conducted by the MHLW.¹³

SLBE: The SLBE is nationwide long-term care claims data. The SLBE covers all residents in Japan as long as they are eligible for LTCI.¹⁴ This database is fiscal-based, including monthly-based claims for 144 months, from April 2006 to March 2018.

In this study, this study uses one-year transition probabilities for disability and mortality risks, by age, sex, and the current level of care demanded, among individuals aged 65-94, estimated in Mikoshiba et al. (2023) using the SLBE. Mikoshiba et al. (2023) constructed one-year interval panel data (30,347,066 records) for a cohort from 1912 to 1951 (7,221,142 unique individuals), and subsequently calculated the average transition probabilities for each age, sex, and current care demand level from 2007 to 2018.

CSLC: The CSLC is a nationally representative repeated cross-sectional micro survey of the non-institutionalized population. The CSLC is established to provide comprehensive information on living status. This study uses data from a large-scale survey conducted in 2016 and from two questionnaires: the household and long-term care questionnaires. This is due to the lack of sociodemographic and socioeconomic information in the SLBE.

according to the MHLW.

¹³Further detailed information can be found here: <https://www.mhlw.go.jp/english/database/db-hss/soltcbe.html> (for the SLBE) (Accessed September 13, 2024). <https://www.mhlw.go.jp/english/database/db-hss/csllc-index.html> (for the CSLC) (Accessed September 13, 2024).

¹⁴Note that the SLBE includes recipients of the welfare transfer program who are eligible for free long-term care assistance. According to the National Survey on Public Assistance Recipients by the MHLW in 2020, of the LTCI recipients in 2020 (about 5.67 million), the recipients of the welfare transfer program account for approximately 6.3% (about 0.32 million).

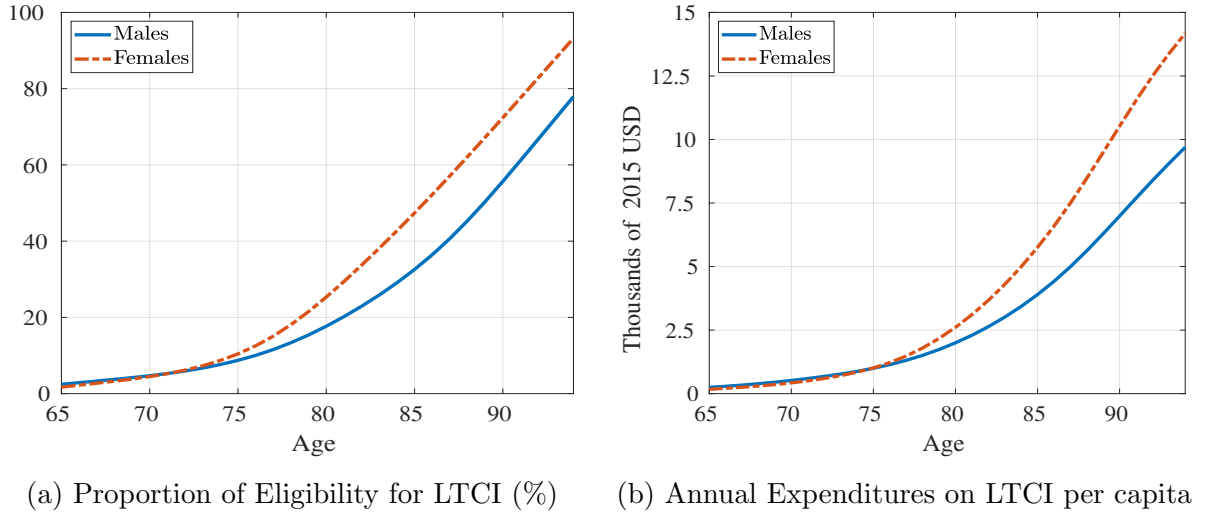


Figure 1: Proportion of Eligibility and Annual Expenditures for LTCI

Note: The data in Figure 1a are from the Report Survey on Situation of Long-term Care Insurance Service by the MHLW in 2015 and the Population Statistics of Japan by the National Institute of Population and Social Security Research (NIPSSR) in 2017. The data in Figure 1b are obtained from the SLBE by the MHLW in 2015 and the Population Statistics of Japan by the NIPSSR in 2017. Annual expenditures are expressed in 1000s of year-2015 dollars using the USD/JPY exchange rate. https://www.stat-search.boj.or.jp/ssi/mtshtml/fm08_m_1.html (in Japanese) (Accessed September 13, 2024).

This study uses the CSLC to capture patterns of care arrangement among older adults with disabilities.

3.2 Long-Term Care Risk in Old Age

This section describes the profiles of the disability and mortality risks in old age. Figure 1 shows the average number of older adults eligible for LTCI services and the average annual gross long-term care expenditures per capita by age and sex in 2015.

3.2.1 Average Disability Risks

As shown in Figure 1a, the average proportion of eligible LTCI recipients increases nearly monotonically with age. On average, the proportion of eligibility is relatively low and remains less than 10% until the mid-70s for both males and females. However, the average proportion grows sharply from their mid-70s and reaches 77.89% and 93.22% at 94 years old for males and females, respectively. Until the mid-70s, the proportion is slightly higher for males than females but subsequently reverses after the mid-70s, with the proportion for females greatly exceeding that for males. From Figure 1b, it is apparent that the average annual gross long-term care expenditures per capita increase with the average proportion of eligibility, particularly after the mid-70s.

3.2.2 Heterogeneity of Disability Risks

Although these average profiles provide suitable information on the expected disability risks and costs in old age, they do not provide information on the heterogeneity of individual risks. Therefore, this study uses transition probabilities by age, sex, and current level of care needs estimated by Mikoshiba et al. (2023). To visualize the dispersion of risks, Mikoshiba et al. (2023) classified long-term care status (LTC-status) into four categories based on the eight levels of care needs: no-disability if independent or ineligible for LTCI; light if the levels range from SL1 to CL2; heavy if the levels range from CL3 to CL5; and death if deceased.

This estimated transition probabilities show significant heterogeneity in the risks of disability and mortality in old age across different ages, sexes, and LTC-statuses. However, they also indicate distinct trajectory patterns in the transitions. First, the disability risks are highly persistent and almost irreversible regardless of age or sex. Individuals of any current LTC-status will most likely remain in the same status in the next year for all ages and sexes. Additionally, once individuals become disabled and eligible for LTCI, they require continuous care until death. The probability of transitioning to the no-disability status is nearly zero for all ages and sexes. Second, there are different risks of disability and mortality for males and females. For all ages, females are more likely to remain in the same LTC-status in the next year than males. However, males have higher mortality rates than females for all ages and LTC- statuses. As shown in Figure 1a, the proportion of eligibility is higher for females than males. This can be explained by the fact that mortality rates are higher for males than for females across different ages and LTC-statuses.

3.3 Care Arrangements

In this section, I describe the care arrangements patterns of older adults with disabilities.

3.3.1 The Breakdown of Care Arrangements

Table 2 provides a detailed breakdown of residential arrangements, calculating the proportion of recipients at home and in nursing homes to the total number of recipients by using the SLBE.¹⁵ As shown in Table 2, the majority of recipients (about 81.82%) receive care services at home. Although only about 18.18% of the recipients use nursing homes, the proportion rises from 4.91% for those with light LTC-status to 38.97% for those with

¹⁵For calculation, this study uses the SLBE, which offers data on the number of recipients at home and in nursing homes because the CSLC does not include information on the institutionalized population. The recipients in nursing homes consist of those in welfare care facilities for the elderly (*tokubetsu-yogo-rojin-home*), healthcare facilities for the elderly (*rojin-hoken-shisetsu*), and nursing care medical facilities(*kaigo-ryoyogata-shisetsu*).

Table 2: Care Arrangements of Older Adults with Disabilities

	At-home care				Public nursing homes
	Care arrangements at home (% at home)				All
	All	Only IC	Mix IC-FHC	Only FHC	
Those eligible for LTCI	81.82%	16.72% (20.44%)	56.22% (68.71%)	8.88% (10.85%)	18.18%
<i>By LTC-status</i>					
Light (SL1-CL2)	95.09%	21.60% (22.72%)	63.26% (66.53%)	10.23% (10.75%)	4.91%
Heavy (CL3-CL5)	61.03%	8.84% (14.48%)	45.41% (74.41%)	6.78% (11.11%)	38.97%

Note: The numbers in the table represent % of all cases. The data are from the SLBE and the CSLC by the MHLW in 2016. The CSLC sample is constructed by matching household questionnaires with long-term care questionnaires, including those eligible for LTCI (65 to 94 years old), with information on the care level, primary and other caregivers, and the FHC service use (5,145 observations). I classify care arrangements based on information on caregivers and the FHC service use. The numbers in the table are derived from the author’s calculation and may not correspond to the numbers published by the MHLW.

heavy LTC-status. This increase can be attributed to the regulations of Japan’s LTCI system, in which older adults with light LTC-status are ineligible for the primary public NHC.

To document facts about care arrangements at home, this study uses the CSLC and construct a sample (the “care sample”) from the CSLC by matching household questionnaires with long-term care questionnaires and comprising individuals eligible for LTCI services, aged between 65 and 94, and with information on the care level and primary caregivers (see the appendix A.1 for more details). Table 2 shows the breakdown of at-home care according to three types of care provision—only IC, mixed use of IC and FHC, and only FHC. I classify at-home care arrangements based on information on relationships with caregivers and the use of FHC services: receiving IC services if receiving care from their own child, child-in-law, spouse and others; and receiving FHC services if using FHC services.

Looking at Table 2, it is apparent that a substantial number of older adults use both IC and FHC at home, accounting for 56.22% of care arrangements (68.71 % of at-home care arrangements). Compared to the U.S. and European countries, the proportion of combining IC and FHC services in Japan stands considerably higher. Barczyk and Kredler (2019) reveal that around 20% of individuals receive both IC and FHC services at home but the contribution of IC to total care hours depends on the public long-term care policy. Japan’s LTCI system provides a benefit-in-kind policy that emphasizes at-home care rather than nursing homes, which may explain the prevalent use of at-home care, particularly the combined usage of IC and FHC services at home.

Table 3: The Burden on Caregiving at Home

	Distr. of primary caregivers			Care hours	The intensity of care by primary caregivers
	(% of all cases)			(annual hrs)	(out of all care hours)
	IC(child)	IC(spouse)	FHC		
Those eligible for LTCI	50.68%	30.96%	14.29%	1904.00	68.59%
<i>By LTC-status</i>					
Light (SL1-CL2)	52.49%	29.24%	14.00%	1424.83	70.00%
Heavy (CL3-CL5)	45.94%	35.44%	15.07%	3094.82	65.15%

Note: The data are from the CSLC by the MHLW in 2016. The CSLC sample is constructed by matching household questionnaires with long-term care questionnaires, including those eligible for LTCI (65 to 94 years old). Column *Distr. of primary caregivers*: The sample is limited to observations with information on the level of care, primary caregivers, other caregivers and the use of FHC services (5,145 observations). The primary caregivers at home is classified into four groups: children if they receive IC from their own child or child-in-law; spouses if receive IC from their spouse; others if receive IC from other family members or others; and FHC if receive FHC. Column *Care hours*: I limit the sample of first-third column to observations with information on the time of the IC and FHC services and monthly expenditures of the FHC services (4,343 observations). Annual total care hours are the sum of annual IC and FHC hours, computing the average annual FHC hours by the total expenditures and the average cost of FHC per hour. Column *Hours by primary caregivers*: I limit the sample for fourth column to observations with positive annual total care hours (4,262 observations). The numbers in the table are derived from the author's calculation and may not correspond to the numbers published by the MHLW.

Another important observation about at-home care from Table 2 is that the proportion of older adults receiving both IC and FHC services (within at-home care) increases with the severity of care. This suggests that the complementarity between IC and FHC becomes more apparent as the level of disability rises, as pointed out by [Bonsang \(2009\)](#). This diminishing trend of the substitution effect may be explained by the higher burden of caregiving and the higher degree of expertise required. For example, regarding the burden, it is difficult to entrust all at-home care, from getting up in the morning to going to bed, to the FHC caregivers from the market.

3.3.2 The Burden of Caregiving at Home

Table 3 also displays the burden of caregiving at home. When looking at primary caregivers, Table 3 shows the important role of family members in providing at-home care. In particular, children and spouses account for about 81.64% of the primary caregivers. In contrast, only 14.29% of the sample receive FHC as their primary caregivers.

The fourth column shows annual care hours provided to older adults with disabilities. There is a notable increase in the annual total care hours as the required level of care increases. The total care hours in heavy LTC-status are approximately double those observed in cases of light LTC-status. This suggests that families with a member in need of severe care, whether in terms of time or financial resources, bear a substantial burden.

In the last column, Table 3 focuses on the intensity of care provided by the primary

caregivers. The last column shows that primary caregivers provide approximately 68.59% of total care hours. Taking into account that the primary caregivers are the family caregivers, this indicates that caregiving burdens are concentrated on family caregivers.

3.3.3 Care Arrangements at Home by Family Structure

IC is the most preferred care option. However, its availability heavily depends on family structure. To be more precise, it relies on the presence of individuals who can provide IC. For instance, for childless older adults, child-provided IC is not an option. Even when older adults have at least one child, they cannot receive IC if their children live far away. Moreover, for married older adults, spouse-provided IC is an available option.

This study then examines the presence of spouses and children among the care sample of the CSLC.¹⁶ Regarding marital status, approximately 95% of the sample is comprised of either widowed individuals (51%) or married individuals (45%). Concerning children, around 91% of the sample has at least one child and approximately 87% of them have at least one child residing together or in the same municipality.

Furthermore, to describe the details of at-home care arrangements by family structure, this study narrows down the care sample to those who are widowed or married, with at least one child residing with them or in the same municipality (the “family sample”). The family sample accounts for 71.26% of the care sample. More than half of the family sample consists of widowed females, making up about 50.76% of the total. Married males, married females, and widowed males constitute 24.86%, 16.74%, and 7.91% of the family sample, respectively. Figure 2 shows the distribution of the family sample by five-year age group, sex, and marital status.

¹⁶For more details, see the appendix A.1.

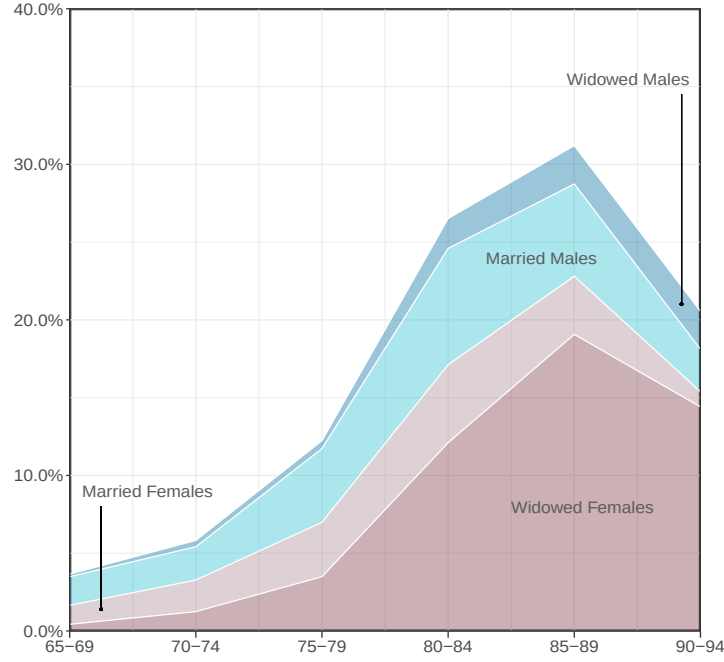


Figure 2: Distribution of “family sample” by five-year age-group, marital status and sex

Note: The data are from the CSLC by MHLW in 2016. The CSLC sample is constructed by matching household questionnaires with long-term care questionnaires, including those who are eligible for LTCI (65 to 94 years old) and those who are widowed or married with at least one child living together or in the same city. Moreover, I limit the CSLC sample to observations with information on the care level, primary and other caregivers (3,692 observations). The numbers in the table are derived from the author’s calculation and may not correspond to the numbers published by the MHLW.

As shown in Figure 2, until their late-70s, married males account for the largest share of the age group. However, after the late-70s, the proportion of widowed females increases sharply and exceeds that of married males. This is consistent with the empirical facts in section 3.2 that the eligibility ratio is slightly higher for males until the mid-70s but reverses later, and males have higher mortality rates than females for all ages and LTC-statuses. Moreover, the number of observations in the family sample increases with age until the early-90s. This is consistent with the empirical facts in section 3.2 that the disability risks increase with age. The fewer number of observations in the 90–94 age group could be explained by the increase in mortality risks with age.

Table 4 presents the care arrangements for widowed females and married males, who make up the majority (75.62%) of the family sample.¹⁷ It can be seen from the data in Table 4 that about 90% of widowed females have their children as primary caregivers. Regarding the characteristics of the children who are primary caregivers, they are of working-age (average age is 58.67 years) and predominantly female (67.63%). Table 4 also shows that about 68.09% of primary caregivers for married males are their wives. The spouses of married males have already retired from the labor market: their average

¹⁷I will present more details of the distribution and characteristics of primary caregivers for the four groups by sex and marital status, including married females and widowed males, in the appendix A.2.

Table 4: The Burden on Caregiving at Home for Widowed Females and Married Males, Having at least one child living together or in the same municipality

	Distr. of primary caregivers			The care intensity	Care arrangements		
	(% of all cases)			by primary caregivers	(% of all cases)		
	IC(child)	IC(spouse)	FHC	(out of all care hrs)	Only IC	Mix IC-FHC	Only FHC
<i><u>Widowed Females</u></i>							
Those elidible for LTCI	88.40%	0.00%	9.71%	65.11%	18.06%	75.42%	6.52%
<i>By LTC-status</i>							
Light (SL1-CL2)	90.21%	0.00%	7.60%	65.83%	20.87%	74.49%	4.64%
Heavy (CL3-CL5)	83.30%	0.00%	15.70%	63.27%	10.10%	78.05%	11.85%
<i><u>Married Males</u></i>							
Those elidible for LTCI	26.14%	68.09%	5.05%	66.19%	26.25%	71.30%	2.45%
<i>By LTC-status</i>							
Light (SL1-CL2)	27.34%	67.81%	4.07%	68.12%	28.84%	68.50%	2.65%
Heavy (CL3-CL5)	23.97%	68.58%	6.81%	62.55%	21.57%	76.35%	2.08%

Note: The data are from the CSLC by MHLW in 2016. The CSLC sample is constructed by matching household questionnaires with long-term care questionnaires, including those who are eligible for LTCI (65 to 94 years old) and those who are widowed or married with at least one child living together or in the same city. Column *Distr. of primary caregivers*: The sample is limited to observations with information on the level of care, primary caregivers, other caregivers and the use of FHC services (1,865 and 915 observations for widowed females and married males, respectively). Column *Hours by primary caregivers*: I limit the sample of first-third column to observations with information on the time of the IC and FHC services and monthly expenditures of the FHC services (1,552 and 783 observations for widowed females and married males, respectively). Column *Care arrangements*: this study uses the sample of first-third column. The numbers in the table are derived from the author's calculation and may not correspond to the numbers published by the MHLW.

age is 76.67 years, and they have lower labor force participation rates (11.86% versus 17.46% among all in their 70s). Moreover, regardless of marital status, sex, and LTC-status, primary caregivers provide high care intensity to recipients, while about 70% of recipients use IC and FHC services at home. These trends for widowed females and married males are also observed for widowed males and married females, respectively.

From these empirical findings, the model in this study focuses primarily on widowed females and their working-age female children. It is important to analyze how disability risks and LTCI affect the labor supply of working-age children under population aging with the rapid decline in the labor force and rising fiscal burden. Furthermore, this study also captures the essence of married males whose caregivers are their wives who have already retired from the labor market.

4 Model

In this section, I build a rich overlapping generations model with two-sided altruism. Families comprises two generations and each generation exhibits altruism toward the other. This study quantifies the effects of long-term care risks on individuals' life-cycle savings and female adult children labor supply through family care arrangements. I

further evaluate the roles of LTCI. The sources of uncertainty in this model are disability and longevity risk in old age, permanent skill shocks, and idiosyncratic wage shocks for females during working-age. The LTC-status affects the total time requirement for care, disability and mortality risk in the next periods, and the composition of the older parent generation. There is no insurance market for risk: there are uninsurable idiosyncratic risks and individuals face a no-borrowing constraint. This is a partial equilibrium model, assuming that individuals take as given the paths of factor prices and various social security policies. The model time is discrete and frequency is annual.

4.1 Demographics

The formalization of the family follows [Fuster et al. \(2007\)](#), [İmrohoroglu and Zhao \(2018\)](#), and [Gao \(2020\)](#), in the sense of two-sided altruism.¹⁸ In this setting, living family members make a joint decision to maximize the same objective functions.

There is a dynastic framework with two stages: an adult child stage and an older parent stage. An individual lives as adult child during the first J periods. At age $J + 1$, they become older parent in the next-generation household of the dynasty. At this time, they leave the labor force. From age $J + 1$, they face disability and mortality risk in each period. The maximum possible age is $2J$.

A family is made up of two generations: an adult child generation (indexed by k) of age $j^k \in \{1, \dots, J\}$ and an older parent generation (indexed by p) of age $j^p = j^k + J$. An individual's life overlaps with their older parent generation household during the first J periods and their adult child generation household in the last J periods. In each family, an older parent generation consists of one household, while an adult child generation consists a household with measure $(1 + \nu)$. The annual population growth rate is v_g and $\nu = (1 + v_g)^J - 1$. In particular, a new generation in a family line is born only in every J period, while a new generation in the economy is born in every period.

During the initial period of each family ($j^k = 1$), each generation household consists of one married couple. Then, each family comprises four individuals from two generations of household, that is, household members in the family $i \in \{kf, km, pf, pm\}$, each representing a female adult child, a male adult child, a female older parent, and a male older parent.¹⁹ While an adult child generation household has two members during J periods, an older parent generation household has $n^p \in [1, 2]$ members.²⁰ A detailed explanation

¹⁸Previous empirical facts on bequest motive and children's help with long-term care are consistent with dynastic motive (two-sided altruism). For example, [Hamaaki et al. \(2019\)](#) find that older parents give a larger share to individuals of the family line and/or those who provide long-term care to their parents. I discuss the modeling choice of two-sided altruism in the appendix [B](#).

¹⁹The average completed number of children per married couple is stable at around 2.2 from 1970 to the early 2000s and drops to 1.94 in 2015, according to the Annual Population and Social Security Surveys (The National Fertility Survey) from the NIPSSR.

²⁰This study does not consider the risk of longevity in middle adulthood because the mortality rates for

of the household members in the older parent generation household is presented in section 4.3. To simplify the model, this study assumes that each household member within the same generation household has the same age and skill.²¹ This study does not model the marriage decision and abstracts from divorce and remarriage.

4.2 Skill

Individuals differ by skill $z = \{L, H\}$ and can be classified as low- and high-skilled. I define individuals as high-skilled if they have a college degree or higher and low-skilled otherwise. At birth, each individual stochastically inherits a skill z from their parents. Individuals' skill state z is fixed throughout the life-cycle and affects their age-specific deterministic labor productivity $\epsilon(j^k, z)$. Notably, individuals' permanent lifetime labor efficiency is deterministic within their entire life, whereas their permanent labor productivity is stochastic between parents and children. z follows a first-order Markov chain of two states with transition probabilities $\Omega(z' | z)$.

Since household members within the same generation have the same skills, four family types can be derived with skill combinations of older parents and adult children. Families of the type in which both parents and children are high-skilled are denoted as HH. The remaining three types are denoted HL, LL, and LH, with the first letter denoting parental skills and the second denoting children's skills.

4.3 Long-Term Care and Mortality Risk

At the beginning of each period, individuals in the older parent generation face disability and mortality risks. In this study, disability and mortality risks are summarized in h , denoting each individuals' LTC-status. This study classifies LTC-status h into four categories based on the eight levels of care needs: no-disability ($h = 1$) if independent or ineligible for LTCI; light ($h = 2$) if the levels range from SL1 to CL2; heavy ($h = 3$) if the levels range from CL3 to CL5; and death ($h = 4$) if deceased.

This model only considers the LTC-status of females in the older parent generation to focus primarily on care arrangements between widowed females and their working-age female adult children—the most common family structure and care arrangements for older adults with disabilities—while keeping the dimensions of the state space manageable.

those 35–65 years are quite low, at 0.26%, according to the Japanese Mortality Database of the NIPSSR (Downloaded on June 27, 2022).

²¹The average age difference between couples from 1975 to 2015 is 2.4 years. The data come from the Vital Statistics conducted by the MHLW. Moreover, there is a high degree of sorting for married couples in Japan. For further details, see Fukuda et al. (2021), which uses the Census data between 1980 and 2010.

4.3.1 Transition of Long-Term Care Status

The LTC-status of the female older parents in the next periods h' depends on her current LTC-status h and age j^p . The LTC-status h follows a first-order Markov chain with transition probabilities $\Psi(h' | h, j^p)$ of being LTC-status h' from age j^p to $j^p + 1$. The death state is the absorbing state for all ages j^p , $\Psi(h' = 4 | h = 4, j^p) = 1$. For simplicity, this study assumes that once individuals become disabled and eligible for LTCI, the probability of transitioning to no-disability status is zero for all ages, because [Mikoshiba et al. \(2023\)](#) show extremely low probabilities (i.e., close to zero).

$$\Psi(h' = 1 | h = 2, j^p) = \Psi(h' = 1 | h = 3, j^p) = 0 \text{ for all } j^p$$

I also assume that the types of care services do not affect the mortality risk, as shown in [Applebaum et al. \(1988\)](#).

4.3.2 Composition of Older Parent Generation

This model also captures quantitatively important aspects of the risks of disability and mortality for males in the older parent generation by assuming that the LTC-status of females affects the composition of the older parent generation as in [Barczyk and Kredler \(2017\)](#). The older parent generation household has $n^p(j^p, h) \in [1, 2]$ members.

No-disability ($h = 1$): When female members in the older parent generation are independent or ineligible for LTCI, I assume that there are $n^p(j^p, h) \in [1, 2]$ members in the older parent generation, consisting of one female and a male member of measure $n^p(j^p, h) - 1 \in [0, 1]$. The measure of the males decreases deterministically with age j^p . This assumption comes from the empirical fact that males have higher mortality rates than females for all ages. In addition, I assume that males are at deterministic risks of disability when females are independent or ineligible for LTCI. Therefore, males receive IC from their wife and pay the average out-of-pocket long-term care expenditures $H_{pm}(j^p)$ to receive FHC services. This assumption reflects the empirical fact that males tend to need care first compared to females and that their primary caregivers are their wives, who are already retired from the labor market.

Eligible for LTCI ($h = 2$ and 3): In the case in which the females need care, I assume that males die and females become widowed. Then, the household of the older parent generation has only one female, $n^p(j^p, h) = 1$.

Death ($h = 4$): When death shocks hit the female older parents, both the female and male members (if still alive) die. This assumption also comes from the empirical fact that males have higher mortality rates than females for all ages. The older parent generation

household has no members, and the family consists of only one generation, that is, the adult child generation.

4.4 Care Arrangements

When females in the older parent generation are eligible for LTCI ($h = 2$ or $h = 3$), their families make care arrangements from three types of care options: IC from their female adult children, FHC, and public NHC (welfare care facilities for the elderly).²² This study models care arrangements as two-stage decision making.

4.4.1 First Stage: At-home Care versus Public Nursing Homes

In first stage, families make residential arrangements between at-home and nursing home care services. This families' choice is denoted by ι , which can be either at-home care services ($\iota = 0$) or public NHC services ($\iota = 1$). If the families choose public NHC services, they incur cost ξ when entering the public NHC.

It is noted that, as mentioned before, older adults with light LTC-status ($h = 2$) are not eligible to use primary public NHC under Japan's LTCI system. The residential arrangement occurs only for older adults with heavy LTC-status ($h = 3$).

4.4.2 Second Stage: Care Arrangements

At-home Care: If families opt for at-home care services ($\iota = 0$) in the first stage, they must determine both the allocated time for IC ϕ and the expenditures for FHC q to meet the minimum requirement of total care hours $\chi(h)$.

$$A(\theta(h)(q/p)^\rho + (1 - \theta(h))(T(\phi) \times 365)^\rho)^{\frac{1}{\rho}} \geq \chi(h) \quad (1)$$

where A denotes the return to care input hours, p is the price of FHC per hour,²³ $\chi(h)$ is the minimum requirement of total care hours depending on the level of care ($\chi(h = 2) \leq \chi(h = 3)$), ρ and $\theta(h)$ represent substitutability between IC and FHC. This formalization captures the heterogeneity of the care burden and the substitutability between IC and

²²This study does not consider private nursing homes as care options. Private nursing homes include fee-based homes for the elderly, residences with health and welfare services for the elderly, and group homes. According to the MHLW, public nursing homes account for the majority in terms of capacity, with welfare care facilities for the elderly (*tokubetsu-yogo-rojin-home*) in particular accounting for the highest proportions, although the capacity of private nursing homes has been increasing. In addition, information available on the occupancy rates in private nursing homes is insufficient. Therefore, this study only focuses on public nursing homes.

²³In Japan's LTCI system, the central government established the fees for each LTCI service and revised them every three years. For further details on the fee, see, for example, <https://www.mhlw.go.jp/topics/kaigo/housyu/housyu.html> (in Japanese) (Accessed September 13, 2024).

FHC by the level of care, following the formulation of [Daruich \(2018\)](#) and [Gao \(2020\)](#). The time of use of IC ϕ is a discrete choice, with the corresponding IC hours.

$$T(\phi) = \begin{cases} 8 \text{ hrs per day} & \text{if } \phi = 1 \\ 4 \text{ hrs per day} & \text{if } \phi = 1/2 \\ 1 \text{ hrs per day} & \text{if } \phi = 1/8 \\ 0 \text{ hrs per day} & \text{if } \phi = 0 \end{cases}$$

Moreover, female older parents with disabilities exhibit preferences for IC, ω . If they have a positive preference for IC, the more hours they use IC, the higher the utility they receive, as shown in section 4.6. The demand for IC would exceed the demand for FHC services covered by LTCI if the preference for IC is relatively high. While the cost incur for public nursing homes ξ affects of choice between at-home care and public nursing homes, the preference of the female older parents for IC ω affects the choice between IC and FHC services.

Public Nursing Homes: Once families select public NHC services ($\iota = 1$) in the first stage, female older parents with disabilities spend the rest of their lives in the public NHC. The families are also required to pay both the care services cost $\bar{q}$ and the facility fee $\bar{c}$ until they die.

4.5 Endowments

Individuals in the adult child generation work in the labor market.

Female Adult Children: Female adult children allocate their disposable time to labor supply, leisure, and IC hours if their female older parents need care. At the beginning of each period, female adult children face an idiosyncratic wage shock, $\mu(j)$. The earnings of female adult children are defined as follows.

$$y_{kf}(j, z) = \epsilon(j, z)\mu(j)\frac{1}{\overline{WH}_{kf}(j)} (\overline{DH}_{kf}(j) - \mathbb{1}_{h \in \{2,3\}}T(\phi) - l)$$

where $\epsilon(j, z)$ is deterministic age-specific efficiency, $\overline{WH}_{kf}(j)$ is the average working hour, $\overline{DH}_{kf}(j)$ is disposable time, and l is leisure. The idiosyncratic wage shock $\mu(j)$ follows the autoregressive AR(1) process.

$$\log(\mu(j)) = \Theta \log(\mu(j-1)) + \zeta(j), \quad \zeta(j) \sim N(0, \sigma_\zeta^2)$$

where $\zeta(j)$ is normally distributed with a mean of zero and variance σ_ζ^2 , and $\Theta < 1$ captures the persistence of the shock. This study discretizes this process into a three-state Markov chain using the method of [Tauchen \(1986\)](#). Let $\Lambda(\mu, \mu')$ be the transition matrix of the idiosyncratic wage shocks. It is also assumed that μ at the time of birth is determined by random draw from initial distribution $\bar{\Lambda}(\mu)$.

Male Adult Children: This study considers that male adult children supply labor inelasticity because their average participation in the labor force participation of working-age males are high. For example, labor force of 35–64-year-old married males is approximately 95.3%, according to the 2017 Employment Status Survey of the Ministry of Internal Affairs and Communications (MIC). The earnings of male adult children are denoted by y_{km} , which evolve deterministically throughout the life-cycle and depend on age and skill $y_{km}(j, z)$.

4.6 Preferences

The utility of the family is the sum of the adult child generation's utility u_k and the older parent generation's utility u_p in the sense of two-sided altruism.

Adult Child Generation: An individual in the adult child generation derives utility from their generation's consumption c_k and leisure l_i for $i = \{kf, km\}$. The instantaneous utility of the adult child generation is given as follows.

$$u_k(c_k, l_{kf}) = \frac{(1 + \nu)}{1 - \sigma} \left(\left(\frac{c_k}{(1 + \nu)\eta(2)} \right)^{1-\gamma} \bar{l}_{km}^\gamma \right)^{1-\sigma} + \frac{(1 + \nu)}{1 - \sigma} \left(\left(\frac{c_k}{(1 + \nu)\eta(2)} \right)^{1-\gamma} l_{kf}^\gamma \right)^{1-\sigma}$$

where l_{kf} denotes the leisure of the female adult child, $\bar{l}_{km}$ is a fixed parameter that represents the exogenous leisure time of the male adult child, $\eta(n)$ is the equivalence scale that varies with the family size.

Older Parent Generation: The utility of the older parent generation depends on the their generation's consumption c_p and the time of use of IC ϕ (when the female older parent with disabilities selects at-home care).

$$u_p(c_p, \phi) = \frac{n^p(j^p, h) - 1}{1 - \sigma} \left(\left(\frac{c_p}{\eta(n^p(j^p, h))} \right)^{1-\gamma} \bar{l}_{pm}^\gamma \right)^{1-\sigma} + \frac{1}{1 - \sigma} \left(\left(\frac{c_p}{\eta(n^p(j^p, h))} \right)^{1-\gamma} \bar{l}_{pf}^\gamma \right)^{1-\sigma} + \mathbb{1}_{h \in \{2, 3\}} \mathbb{1}_{\iota=0}(\omega\phi)$$

where ω represents preference parameter for IC. If $\omega > 0$, the female older parent prefers IC to FHC. Note that leisure l_i for $i = \{pf, pm\}$ is the exogenous time of the individual in the older parent generation.

4.7 Government

The government operates the following social insurance programs: LTCI, health insurance, pay-as-you-go public pension, and means-tested welfare transfer program.

4.7.1 Public Long-Term Care Insurance

The government provides the mandatory public LTCI system based on the level of care demanded, regardless of socioeconomic attributes. All individuals 65 years and above receive care services covered by LTCI if they are certified as needing long-term care or support. Out-of-pocket long-term care expenditures paid by each recipient are denoted as H_i^{oop} for $i \in \{pf, pm\}$ and expressed as follows.

$$\begin{aligned} H_{pf}^{oop} &= \lambda^h q \\ H_{pm}^{oop} &= \lambda^h H_{pm} \end{aligned}$$

where λ^h is the copayment rate of the LTCI. The government covers the fraction $(1 - \lambda^h)$ of gross long-term care expenditures.

4.7.2 Public Health Insurance

The government also offers the mandatory public health insurance program. This study assumes that medical expenditures are required when individuals are in the older parent generation.^{24 25} The average annual gross medical expenditures are given exogenously, $M_i(j^p)$ for $i \in \{pf, pm\}$, depending on age and sex. Out-of-pocket medical expenditures are defined similarly to $M_i^{oop} = \lambda_{j^p}^m M_i(j^p)$, where $\lambda_{j^p}^m$ is the copayment rate of the health insurance depending on the age. The government covers the remaining fraction $(1 - \lambda_{j^p}^m)$ of gross medical expenditures.

4.7.3 Public Pension

The government operates a pay-as-you-go public pension system. Individuals receive public pension benefits once they are in the older parent generation. Let $\text{pen}_i(j^p, z)$ denote the public pension benefits for individuals $i \in \{pf, pm\}$ with age j^p and permanent skill z . I assume that the benefits of male older parents are determined as follows.

$$\text{pen}_{pm}(j^p, z) = \kappa \frac{\bar{y}_m(z)}{J - 1} \quad (2)$$

²⁴This study does not consider medical expenditures during the adult child stage. This is because average annual gross medical expenses during the adult child stage are relatively low. According to the National Medical Expenses of the MHLW in 2015, average annual gross medical expenses remain close to 200,000 yen until 50 years, and subsequently increase, but until age 65 they are at most less than 400,000 yen and 500,000 yen for females and males, respectively. Website: https://www.e-stat.go.jp/stat-search/files?stat_infid=000031622557 (in Japanese) (Accessed September 13, 2024).

²⁵This study assumes that no correlation between long-term care and medical expenditures. Suzuki et al. (2012) report no correlation between long-term care and medical expenditures after controlling for inpatients and nursing home residents. This study uses the complete set of insurance claims data provided by public insurers of Fukui Prefecture in Japan.

where κ is the public pension replacement rate and

$$\bar{y}_m = \begin{cases} y_{km}(j, z) & \text{if } j = 1 \\ y_{km}(j, z) + \bar{y}_{km}(j - 1, z) & \text{if } 1 < j \leq J \\ \bar{y}_{km}(j - 1, z) & \text{if } J < j \end{cases}$$

I further assume that the benefits of female older parents depend on the average earnings of the skill group instead of the past individual earnings.²⁶

$$\text{pen}_{pf}(j^p, z) = \kappa \frac{1}{J-1} \sum_{j=1}^J \mathbb{E} \left[\epsilon(j, z) \mu(j) \frac{1}{\overline{WH}_{kf}} (\overline{DH}_{kf} - \mathbb{1}_{h \in \{2,3\}} T(\phi_j) - l_j) \right] \quad (3)$$

4.7.4 Means-tested Welfare Transfer Program

Individuals with low income and savings can be eligible for the means-tested welfare program (i.e., *seikatsu-hogo*). This covers their minimum living expenses and their long-term care and medical expenditures. A means-tested transfer tr is provided to guarantee a minimum consumption level $\underline{c}$ for each household in each generation. The minimum consumption level differs by marital status. The transfer amount for a family is given as follows.

$$tr = \max \left\{ 0, (1 + \tau^c)(\underline{c}_k + \underline{c}_p) - \left(Ra + \sum_{i \in \{kf, km\}} (1 - \tau^l)(1 + \nu)y_i + \sum_{i \in \{pf, pm\}} (\text{pen}_i - M_i^{oop} - H_i^{oop}) \right) \right\}$$

As in [De Nardi et al. \(2010\)](#), this study imposes that if transfers are positive, the family consumes all of its resources, i.e., $a' = 0$.

4.7.5 Taxes

The government imposes proportional taxes on consumption at rate τ^c , labor income at τ^l , capital income at τ^a , and lump-sum tax τ^{ls} on each individual. The net-of-tax gross return on capital is denoted as $R = 1 + (1 - \tau^a)r$, where r is the interest rate. The government budget constraint is given as follows.

$$\tau^l Y_l + \tau^a Y_a + \tau^c (C_k + C_p) + \tau^{ls} N = SS + HI + LTC + TR + G \quad (4)$$

where Y_l, Y_a, C_k , and C_p denote aggregate labor income, capital income, and consumption for the adult child and older parent generation, respectively; N denotes the total number of individuals; SS, HI, LTC , and TR each denote the total government expenditures for a public pension, public health insurance, public LTCI, and the means-tested welfare program, respectively; G denotes the government's consumption expenditures.

²⁶Although public pension benefits depend on past individual earnings in the actual economy, a substantial additional burden for computation arises when a new state variable, such as average lifetime earnings, is introduced. To keep the state space dimensions manageable, this study follows the formalization of [Attanasio et al. \(2010\)](#).

In the baseline model, I assume that τ^{ls} is zero and let G absorb the imbalance and satisfy equation (4) to isolate the effects of governmental long-term care expenditure and focus on changes from different risks individuals face over the life-cycle. In the numerical experiments in section 6, this study considers various policy scenarios and adjust τ^{ls} to account for a change in the net government revenues to balance the government budget in equation (4).

4.8 Problem of Families

Families are heterogeneous in terms of the age of the adult child generation j^k , asset a , skill of the older parent generation z , skill of the adult child generation z' , current LTC-status of the female older parent h , use of NHC for the female older parent in the previous period ι_{-1} , and idiosyncratic wage shock for the female adult child μ . I summarize the state as $\mathbf{x} = \{j^k, a, z, z', h, \iota_{-1}, \mu\}$. I define the problem of families with six value functions, based on the current LTC-status h and previous residential arrangements ι_{-1} of the female older parent.

4.8.1 No Parents ($h = 4$)

The state variables of families without parents are given as $(j^k, a, z, z', h = 4, \mu)$. The value function is expressed as follows.

$$V_{j^k}^K(a, z, z', h = 4, \mu) = \max_{c_k, l_{kf}, a'} \left\{ u_k(c_k, l_{kf}) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') \right\}$$

subject to

$$(1 + \tau^c)c_k + a' = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + tr$$

where

$$y_{kf} = \epsilon \mu \left((\overline{DH}_{kf} - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf}$$

$$a' \geq 0, \quad c_k \geq 0$$

$$\mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) V_{j^k+1}^K(a', z, z', h' = 4, \mu') & \text{if } j^k < J \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.8.2 Heavy LTC-status in Public Nursing Homes ($h = 3$ and $\iota_{-1} = 1$)

The state variables of families are given as $(j^k, a, z, z', h = 3, \iota_{-1} = 1, \mu)$. Note that once families select NHC, their female older parents with disabilities spend the rest of their lives in the public NHC. Their families then pay both the cost of long-term care services

$\bar{q}$ and the fee for the use of the NHC $\bar{c}$, including the residence fee, food fee, and expenses of daily living.

$$V_{j^k}^{HI}(a, z, z', h = 3, \iota_{-1} = 0, \mu) = \max_{c_k, l_{kf}, a'} \left\{ u_k(c_k, l_{kf}) + u_p(\bar{c}) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') \right\}$$

subject to

$$(1 + \tau^c)(c_k + \bar{c}) + a' + H_{pf}^{oop} = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + \text{pen}_{pf} - M_{pf}^{oop} + tr$$

where

$$\begin{aligned} y_{kf} &= \epsilon \mu \left((\overline{DH}_{kf} - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf} \\ H_{pf}^{oop} &= \lambda^h \bar{q} \\ a' &\geq 0, \quad c_k \geq 0 \end{aligned}$$

$$\mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) [\Psi(h' = 2 \mid h = 3, j^p) V_{j^k+1}^{LI}(a', z, z', h' = 2, \iota_{-1} = 1, \mu') \\ \quad + \Psi(h' = 3 \mid h = 3, j^p) V_{j^k+1}^{HI}(a', z, z', h' = 3, \iota_{-1} = 1, \mu') \\ \quad + \Psi(h' = 4 \mid h = 3, j^p) V_{j^k+1}^K(a', z, z', h' = 4, \mu')] & \text{if } j^k < J \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.8.3 Heavy LTC-status at Home ($h = 3$ and $\iota_{-1} = 0$)

The state variables of families are expressed as $(j^k, a, z, z', h = 3, \iota_{-1} = 0, \mu)$. The value function is expressed as follows.

$$\begin{aligned} V_{j^k}^{HC}(a, z, z', h = 3, \iota_{-1} = 0, \mu) \\ = \max_{\iota \in \{0,1\}} \left\{ (1 - \iota) \left(\tilde{V}_{j^k}^{HC}(a, z, z', h = 3, \iota_{-1} = 0, \mu) \right) + \iota \left(V_{j^k}^{HI}(a, z, z', h = 3, \iota_{-1} = 0, \mu) + \xi \right) \right\} \end{aligned}$$

If families choose NHC in the next periods ($\iota = 1$), see section 4.8.2. If families chooses at-home care services ($\iota = 0$), families have to determine both the allocated time for IC ϕ and the expenditures for FHC q to satisfy equation (1). Note that q is determined by minimizing out-of-pocket long-term care expenditures when ϕ is given as follows.

$$q^*(\phi) = \begin{cases} 0 & \text{if } (\chi(h)/A)^\rho - (1 - \theta(h))(T(\phi) \times 365)^\rho \leq 0 \\ \left(\frac{(\chi(h)/A)^\rho - (1 - \theta(h))(T(\phi) \times 365)^\rho}{\theta(h)} \right)^{\frac{1}{\rho}} p & \text{if } (\chi(h)/A)^\rho - (1 - \theta(h))(T(\phi) \times 365)^\rho > 0 \end{cases}$$

Thereafter, the value function can be rewritten as follows.

$$\tilde{V}_{j^k}^{HC}(a, z, z', h = 3, \iota_{-1} = 0, \mu) = \max_{\phi \in \{0, 1/8, 1/2, 1\}} \left\{ \max_{c_k, c_p, l_{kf}, a'} \{ u_k(c_k, l_{kf}) + u_p(c_p, \phi) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') \} \right\}$$

subject to

$$(1 + \tau^c)(c_k + c_p) + a' + H_{pf}^{oop} = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + \text{pen}_{pf} - M_{pf}^{oop} + tr$$

where

$$\begin{aligned} y_{kf} &= \epsilon\mu \left((\overline{DH}_{kf} - T(\phi) - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf} - T(\phi) \\ H_{pf}^{oop} &= \lambda^h q^*(\phi) \\ a' &\geq 0, \quad c_k, c_p \geq 0 \end{aligned}$$

$$\mathbb{E} \tilde{V}_{j^k+1} = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) [\Psi(h' = 2 \mid h = 3, j^p) V_{j^k+1}^{LC}(a', z, z', h' = 2, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 3 \mid h = 3, j^p) V_{j^k+1}^{HC}(a', z, z', h' = 3, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 4 \mid h = 3, j^p) V_{j^k+1}^K(a', z, z', h' = 4, \mu')] & \text{if } j^k < J \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.8.4 Light LTC-status in Public Nursing homes ($h = 2$ and $\iota_{-1} = 1$)

The state variables of families are given as $(j^k, a, z, z', h = 2, \iota_{-1} = 1, \mu)$. The value function is expressed as follows.

$$V_{j^k}^{LI}(a, z, z', h = 2, \iota_{-1} = 0, \mu) = \max_{c_k, l_{kf}, a'} \left\{ u_k(c_k, l_{kf}) + u_p(\bar{c}) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') \right\}$$

subject to

$$(1 + \tau^c)(c_k + \bar{c}) + a' + H_{pf}^{oop} = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + \text{pen}_{pf} - M_{pf}^{oop} + tr$$

where

$$\begin{aligned} y_{kf} &= \epsilon\mu \left((\overline{DH}_{kf} - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf} \\ H_{pf}^{oop} &= \lambda^h \bar{q} \\ a' &\geq 0, \quad c_k \geq 0 \end{aligned}$$

$$\mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) [\Psi(h' = 2 \mid h = 2, j^p) V_{j^k+1}^{LI}(a', z, z', h' = 2, \iota_{-1} = 1, \mu') \\ \quad + \Psi(h' = 3 \mid h = 2, j^p) V_{j^k+1}^{HI}(a', z, z', h' = 3, \iota_{-1} = 1, \mu') \\ \quad + \Psi(h' = 4 \mid h = 2, j^p) V_{j^k+1}^K(a', z, z', h' = 4, \mu')] & \text{if } j^k < J \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.8.5 Light LTC-status at Home ($h = 2$ and $\iota_{-1} = 0$)

The state variables of families are given as $(j^k, a, z, z', h = 2, \iota_{-1} = 0, \mu)$. The value function is expressed in the following way.

$$V_{j^k}^{LC}(a, z, z', h = 2, \iota_{-1} = 0, \mu) = \max_{\phi \in \{0, 1/8, 1/2, 1\}} \left\{ \max_{c_k, c_p, l_{kf}, a'} \{u_k(c_k, l_{kf}) + u_p(c_p, \phi) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}')\} \right\}$$

subject to

$$(1 + \tau^c)(c_k + c_p) + a' + H_{pf}^{oop} = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + \text{pen}_{pf} - M_{pf}^{oop} + tr$$

where

$$y_{kf} = \epsilon \mu \left((\overline{DH}_{kf} - T(\phi) - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf} - T(\phi)$$

$$H_{pf}^{oop} = \lambda^h q^*(\phi)$$

$$a' \geq 0, \quad c_k, c_p \geq 0$$

$$\mathbb{E} \tilde{V}_{j^k+1} = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) [\Psi(h' = 2 \mid h = 2, j^p) V_{j^k+1}^{LC}(a', z, z', h' = 2, \iota_{-1} = 0, \mu') & \text{if } j^k < J \\ \quad + \Psi(h' = 3 \mid h = 2, j^p) V_{j^k+1}^{HC}(a', z, z', h' = 3, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 4 \mid h = 2, j^p) V_{j^k+1}^K(a', z, z', h' = 4, \mu')] \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.8.6 No-Disability ($h = 1$)

The state variables of families are given as $(j^k, a, z, z', h = 1, \iota_{-1} = 0, \mu)$. The value function is expressed in the following way.

$$V_{j^k}^{ND}(a, z, z', h = 1, \iota_{-1} = 0, \mu) = \max_{c_k, c_p, l_{kf}, a'} \left\{ u_k(c_k, l_{kf}) + u_p(c_p) + \beta \mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') \right\}$$

subject to

$$\begin{aligned} & (1 + \tau^c)(c_k + c_p) + a' \\ & = Ra + (1 - \tau^l)(1 + \nu)(y_{kf} + y_{km}) + \sum_{i \in \{pf, pm\}} \text{pen}_i - \sum_{i \in \{pf, pm\}} M_i^{oop} - H_{pm}^{oop} + tr \end{aligned}$$

where

$$y_{kf} = \epsilon \mu \left((\overline{DH}_{kf} - l_{kf}) / \overline{WH}_{kf} \right), \quad 0 \leq l_{kf} \leq \overline{DH}_{kf}$$

$$a' \geq 0, \quad c_k, c_p \geq 0$$

$$\mathbb{E} \tilde{V}_{j^k+1}(\mathbf{x}') = \begin{cases} \sum_{\mu'} \Lambda(\mu', \mu) [\Psi(h' = 1 \mid h = 1, j^p) V_{j^k+1}^{ND}(a', z, z', h' = 1, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 2 \mid h = 1, j^p) V_{j^k+1}^{LC}(a', z, z', h' = 2, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 3 \mid h = 1, j^p) V_{j^k+1}^{HC}(a', z, z', h' = 3, \iota_{-1} = 0, \mu') \\ \quad + \Psi(h' = 4 \mid h = 1, j^p) V_{j^k+1}^K(a', z, z', h' = 4, \mu')] & \text{if } j^k < J \\ (1 + \nu) \sum_{z''} \Omega_{z''|z'} \sum_{\mu'} \bar{\Lambda}(\mu') V_1^{ND} \left(\frac{a'}{(1+\nu)}, z', z'', h' = 1, \iota_{-1} = 0, \mu' \right) & \text{if } j^k = J \end{cases}$$

4.9 Equilibrium

Given the interest rate r , and a set of government policies $\{\lambda^h, \lambda^m, \tau^c, \tau^a, \tau^l\}$, a stationary recursive competitive equilibrium is a set of six value functions $\{V_{j^k}^{ND}(\mathbf{x}), V_{j^k}^{LC}(\mathbf{x}), V_{j^k}^{LI}(\mathbf{x}), V_{j^k}^{HC}(\mathbf{x}), V_{j^k}^{HI}(\mathbf{x}), V_{j^k}^K(\mathbf{x})\}_{j^k=1}^J$, family decision rules $\{c_{k,j^k}(\mathbf{x}), c_{p,j^k}(\mathbf{x}), l_{kf,j^k}(\mathbf{x}), a_{j^k+1}(\mathbf{x}), \iota_{j^k}(\mathbf{x}), \phi_{j^k}(\mathbf{x}), q_{j^k}(\mathbf{x})\}_{j^k=1}^J$, time-invariant measures of families $X_{j^k}(\mathbf{x})$ with age- j^k families with the state vector $\mathbf{x} = \{a, z, z', h, \iota_{-1}, \mu\}$, and lump-sum transfer τ^{ls} , such that the following conditions are satisfied. See the appendix C for further details of the numerical procedures.

1. Given the factor prices and government policies, the family decision rules solve the family decision problems in section 4.8.
2. The government budget is balanced in equation (4).
3. Individuals and aggregate behaviors are consistent as follows.

$$\begin{aligned} Y_l &= \sum_{j^k=1}^J \sum_{\mathbf{x}} [y_{kf}(\mathbf{x}) + y_{km}(\mathbf{x})] X_{j^k}(\mathbf{x}) \\ &= \sum_{j^k=1}^J \sum_{\mathbf{x}} \left[\frac{\epsilon(j^k, z') \mu(j^k)}{\bar{W} \bar{H}_{kf}(j^k)} (\bar{D} \bar{H}_{kf}(j^k) - \mathbb{1}_{h \in \{2,3\}} T(\phi_{j^k}) - l_{kf,j^k}(\mathbf{x})) + y_{km}(j^k, z') \right] X_{j^k}(\mathbf{x}) \\ Y_a &= \sum_{j^k=1}^J \sum_{\mathbf{x}} R a_{j^k}(\mathbf{x}) X_{j^k}(\mathbf{x}) \\ C_k &= \sum_{j^k=1}^J \sum_{\mathbf{x}} c_{k,j^k}(\mathbf{x}) X_{j^k}(\mathbf{x}) \\ C_p &= \sum_{j^k=1}^J \sum_{\mathbf{x}} c_{p,j^k}(\mathbf{x}) X_{j^k}(\mathbf{x}) \\ N &= \sum_{j^k=1}^J \sum_{\mathbf{x}} [2 + n^p(j^p, h)] X_{j^k}(\mathbf{x}) \\ SS &= \sum_{j^k=1}^J \sum_{\mathbf{x}} [\text{pen}_{pf}(j^p, z') + (n^p(j^p, h) - 1) \text{pen}_{pm}(j^p, z')] X_{j^k}(\mathbf{x}) \end{aligned}$$

$$\begin{aligned}
HI &= \sum_{j^k=1}^J \sum_{\mathbf{x}} (1 - \lambda^m(j^p)) [M_{pf}(j^p) + (n^p(j^p, h) - 1)M_{pm}(j^p)] X_{j^k}(\mathbf{x}) \\
LTC &= \sum_{j^k=1}^J \sum_{\mathbf{x}} (1 - \lambda^h) [q_{j^k}^*(\mathbf{x}) + (n^p(j^p, h) - 1)H_{pm}(j^p)] X_{j^k}(\mathbf{x}) \\
TR &= \sum_{j^k=1}^J \sum_{\mathbf{x}} tr(\mathbf{x}) X_{j^k}(\mathbf{x})
\end{aligned}$$

4. The public pension benefit system is balanced in equations (2) and (3).
5. The set of age-dependent measures of families satisfies the following conditions.
 - For $j^k < J$,

$$\begin{aligned}
&X_{j^k+1}(a', z, z', h', \iota, \mu') \\
&= \frac{1}{(1 + \nu)^{1/J}} \sum_{\{a, h, \iota_{-1}, \mu: a', \iota\}} \Psi(h' | h, j^p) \Lambda(\mu', \mu) X_{j^k}(a, z, z', h, \iota_{-1}, \mu) \quad (5)
\end{aligned}$$

where a' and ι are the optimal choices in the later periods.

- For $j^k = J$,

$$\begin{aligned}
&X_1(a', z', z'', h' = 1, \iota = 0, \mu') \\
&= (1 + \nu) \sum_{\{a, z, h, \iota_{-1}, \mu: a'\}} \Omega_{z''|z'} \bar{\Lambda}(\mu') X_J(a, z, z', h, \iota_{-1}, \mu) \quad (6)
\end{aligned}$$

where a' is the optimal choice in the next periods.

5 Calibration

This section describes the calibration of the model parameters. I calibrate the steady state economy to the Japanese economy in 2015. The parameters in this model are divided into two categories. In the first category, those are external parameters directly estimated from the data and existing literature. The parameters in the second group are internal parameters calibrated by matching the model-generated targets' values to their data counterparts.

5.1 External Calibration

I turn to parameters that I estimate directly from the data and existing literature. The parameters and values are summarized in Table 5.

5.1.1 Demographics

I let individuals enter the economy at age $j = 1$, corresponding to 35 years old. I set the age difference between an adult child generation and an older parent generation as 30 years because the average age difference between mothers and children from 1975–2015 is 30.05 years according to the Vital Statistics of the MHLW in 2019. Further, individuals retire from the labor market at 65 years and live to the maximum possible age of 94. I set the annual population growth rate at zero. The equivalence scale η adjusts the consumption of each generation according to the size of the household, which assigns $\eta(n) = 1 + 0.7(n - 1)$ to the size of the family n , following [Bick and Choi \(2013\)](#).

5.1.2 Long-Term Care Risk and Medical Expenditure Risk

I use transition probabilities Ψ by age, sex, and current level of care needs estimated in [Mikoshiha et al. \(2023\)](#), described in section 3.2. I assume that the number of household members in the older parent generation $n^p(j^p, h)$ depends on both the age and LTC-status discussed in section 4.3. The deterministic measure of the male older parent $n^p(j^p, h = 1) - 1$ is calibrated based on their survival probabilities, which are estimated by [Mikoshiha et al. \(2023\)](#).

As shown in Figure 1b, I calculate the average annual gross long-term care expenditures per capita for male older parents H_{pm} from the SLBE of the MHLW in 2015 and the Population Statistics of Japan of the NIPSSR in 2017.²⁷ Further, I calculate the average annual gross medical expenditures for older parents $M_i(j^p)$ for $i \in \{pf, pm\}$ from the National Medical Expenses (NME) of the MHLW in 2015.

5.1.3 Skill

The transition probability of skill inheritance Ω is calibrated to match both the proportion of high-skilled individuals in the working-age population and the correlation between the income of children and parents, as in [İmrohoroglu and Zhao \(2018\)](#). The proportion of high-skilled individuals is 31%, as reported by the Employment Status Survey (ESS) of the MIC in 2017. I use the estimated value of the correlation between children’s and parents’ income from [Lefranc et al. \(2014\)](#). The transition probabilities of skill inheritance are given as follows.

$$\Omega = \begin{bmatrix} \Omega_{LL} & \Omega_{LH} \\ \Omega_{HL} & \Omega_{HH} \end{bmatrix} = \begin{bmatrix} 0.80 & 0.20 \\ 0.45 & 0.55 \end{bmatrix}$$

²⁷Note that the Population Statistics of Japan 2017 provides the annual population by age and sex in 2015. Website: <https://www.ipss.go.jp/syoushika/tohkei/Popular/Popular2017RE.asp?chap=0> (Accessed September 13, 2024).

where the generic element $\Omega_{zz'}$ with $z, z' \in \{L, H\}$ is the probability of the transition of inherited skills from the older parent generation with skill z to the adult child generation with skill z' . In the steady state, the distribution of skill combinations between the older parent generation and the adult child generation is 54.90%, 13.95%, 13.95%, and 17.21% for LL, HL, LH, and HH, respectively.

5.1.4 Endowments

The age-specific deterministic labor productivity $\epsilon(j^k, z)$ for the working-age married females is calibrated from their earnings based on the ESS of the MIC. I use the 2017 ESS data and adjust them to the 2015 level using the Consumer Price Index (CPI). Figure 3 shows the life-cycle earnings profiles for the working-age married females by age and skill to calibrate their labor productivity. Figure 3 shows that high-skilled married females earn more over the working-age than low-skilled married. As well documented in studies such as Kitao and Mikoshiba (2020) and Kitao and Mikoshiba (2022), many female workers leave the labor force at child-bearing age and return to work after several years, resulting in the so-called “M-shaped” patterns.

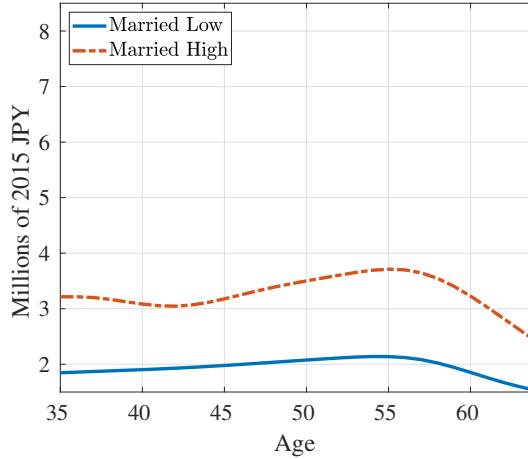


Figure 3: Earnings of Married Females by Age and Skill

Note: Figure 3 shows married females’ earnings by age and skill. The married sample includes both widowed and divorced individuals. I define individuals as high-skilled if they have a college or higher degree and low-skilled otherwise. The data are obtained from the Employment Status Survey (ESS) of the Ministry of Internal Affairs and Communications (MIC) in 2017. I use the data from 2017 and adjust them to the 2015 level using the consumer price index.

To capture the labor supply at the extensive and intensive margins of married females, this study introduces both average working hours $\overline{WH}_{yf}(j^k)$ and disposable income $\overline{DH}_{yf}(j^k)$ for married females across ages. I normalize disposable time to 1.0 and calibrate the average working hours using the Time Use Survey of the MIC in 2016. For the idiosyncratic wage shock, I discretize the AR(1) process into a three-state Markov chain as in Tauchen (1986). Following Hsu and Yamada (2019), I take $\Theta = 0.98$ and vari-

ance $\sigma_\zeta = 0.09$. The resulting value of μ is $\{0.40, 1.00, 2.47\}$, and the initial distribution $\bar{\Lambda}(\mu)$ is $\{0.21, 0.58, 0.21\}$.

The average earnings of married males $y_{km}(j^k, z)$ vary deterministically with age and skill. I compute them using the ESS data on the average earnings of married males, using the data for 2017 and adjusting them to the 2015 level by the CPI. Figure 4 shows the life-cycle profiles of average earnings of married males by age and skill. As well documented in studies such as Kitao and Mikoshiba (2020), there are large differences in earnings by sexes and skill levels. Regardless of skill level, male earnings are much higher than female earnings. High-skilled married males earn the most.

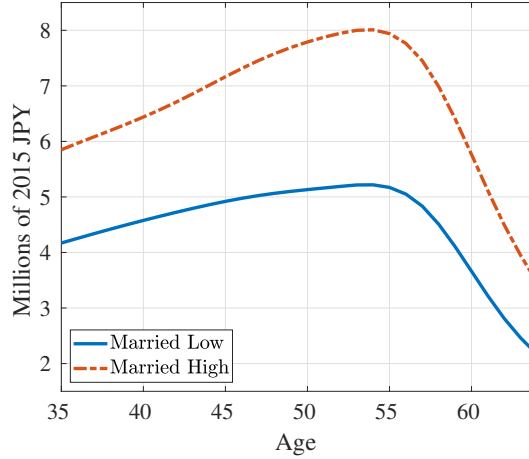


Figure 4: Average Earnings of Married Males by Age and Skill

Note: Figure 4 reveals the average earnings of married males by age and skill. Average earnings are calculated by multiplying earnings by the labor force participation of married males. The married sample includes both widowed and divorced individuals. I define individuals as high-skilled if they have a college or higher degree and as low-skilled otherwise. The data are obtained from the Employment Status Survey (ESS) of the Ministry of Internal Affairs and Communications (MIC) in 2017. I use the data from 2017 and adjust them to the 2015 level using the consumer price index.

The leisure of the male adult child, the male older parent, and the female older parent is calculated using the 2016 Time Use Survey of the MIC. The values obtained are 0.54, 0.54, and 0.50, respectively.

5.1.5 Care Arrangements

For the cost of NHC services, I use the Survey of Institutions and Establishments for Long-term Care (SIEL) of the MHLW in 2016. The SIEL reports the average cost for different types of expenditures for NHC services covered under LTCI by different levels of care. I calculate the weighted average cost of NHC services $\bar{q}$ and the weighted average fee $\bar{c}$ for the welfare care facilities for the elderly (special nursing homes). I set them at 327.83 and 121.20 (10,000-yen, CPI adjusted to 2015), respectively. Since NHC recipients in the facility must pay for their living costs, I calculate the weighted average living costs

$\bar{c}$ as the sum of the residence fee, food fee, and daily living expenses.²⁸

For the parameters of at-home care, the average cost of FHC per hour p and the minimum requirement of total care hours $\chi(h)$ are calibrated directly from the CSLC in 2016, using information on the IC and FHC services hours and monthly expenditures on FHC services for the care sample. I calculate the average cost of FHC per hour p as 0.167 (10,000-yen, CPI adjusted to 2015), using the information on time and monthly expenditures for FHC services. As the minimum requirement of total care hours, I calculate average annual total care hours as 1424.83 and 3094.82 hours for the light and heavy LTC-status, respectively. Annual total care hours are the sum of annual IC and FHC hours, computing the average annual FHC hours by the total expenditures and the average cost of FHC per hour.

5.1.6 Preference

The coefficient of relative risk aversion σ is set at 3.0, which is in the range of values used in the literature. For example, [De Nardi et al. \(2016\)](#) set the risk aversion at 2.83 by the model estimation.

5.1.7 Government

The government operates the public LTCI, public health insurance, pay-as-you-go public pension, and means-tested welfare transfer program. The copayment rate of the LTCI λ^h is set to 10% for all ages. The LTCI covers 90% of the long-term care expenditures for both FHC and public NHC services. The public health insurance also covers part of the medical expenditure, and its copayment rate λ_{jp}^m varies with age. I set λ_{jp}^m at 30%, 20%, and 10% for those aged 69 years and below, between 70 and 74, and above 75, respectively. The pension replacement rate κ is set at one-third, based on the [OECD \(2019\)](#)'s estimated average gross replacement rate of public pensions.

The means-tested welfare program of my model provides means-tested transfers to eligible households. The consumption floor is set at 87 and 132 (10,000-yen) for widowed and married couples, respectively. The amount is set to be within the range of average public assistance payments according to the family size.²⁹

I set the consumption tax rate τ^c at 8% based on the tax rate in 2015. Furthermore,

²⁸The average residence fee is set at 58.83 (10,000-yen) as the average standard amount of residence fee by different types of institutions. The average food fee is set at 50.37 (10,000-yen) by the standard amount of the food fee, and the living costs are set at 12.00 (10,000-yen). For details of the standard amount of living costs, see, for example, <https://www.kaigokensaku.mhlw.go.jp/commentary/fee.html> (in Japanese) (Accessed September 13, 2024).

²⁹For more details on the program, see, for example, https://www.mhlw.go.jp/file/05-Shingikai-12601000-Seisakutoukatsukan-Sanjikanshitsu_Shakaihoshoutantou/kijun23_05.pdf (in Japanese) (Accessed September 13, 2024).

I set the labor tax rate τ^l and capital tax rate τ^a at 30% and 35%, respectively, based on [Gunji and Miyazaki \(2011\)](#) and [Kitao and Mikoshiba \(2020\)](#)—consistent with the literature estimates of effective income tax rates.

5.1.8 Factor Prices

This model is a partial equilibrium model, assuming that the interest rate r is exogenous. I set the interest rate to 2% based on [Aoki et al. \(2016\)](#).

5.2 Internal Calibration

The rest of the parameters to be determined are the distribution of preferences and the care arrangements. I calibrate remaining parameters by matching the model-generated targets to their data counterparts, using the simulated method of moments. This study minimizes a simple residual sum of squares, using a Nelder-Mead method. Table 6 summarizes the description and values of the parameters.

For the subjective discount factor β , I use the data from [Kitao and Yamada \(2019\)](#), who use the National Survey of Family Income and Expenditure (NSFIE) of the MIC in 2014. I set the parameter to ensure that the model achieves the average per adult equivalent wealth of 823.93 (10,000-yen, CPI adjusted to 2015). I set the intensity of leisure in the utility function γ to correspond to the average labor force participation of working-age married males, which is 70.71%. I use the 2017 ESS data provided by the MIC. The calibrated value of γ is in the range of values estimated in the literature, such as [Fuster et al. \(2007\)](#) at 0.63 and [Gao \(2020\)](#) at 0.42.

Next, I turn to calibrate six parameters about care arrangements. The data on care arrangements are obtained from the CSLC in 2016. The average annual long-term care hours help identify the parameter of returns to care input hours A . For the substitution between IC and FHC, I use the correlation between IC and FHC hours ρ and the ratio of IC hours in total care hours by LTC-status $\theta(h)$. For the correlation between IC and FHC hours, I group the eligible older adults by the quartile of FHC hours, as in [Daruich \(2018\)](#). The ratio of those who use only IC is to calibrate preference for IC ω . The preference for NHC ξ is calibrated to the share of nursing home users.

Table 5: Exogenously Calibrated Parameters

Parameter	Description	Values / Source
<i>Demographics</i>		
J	Lifetime span	30
ν_g	Population growth rate	0 %
η	Equivalence scale	Bick and Choi (2013)
<i>Disability and Mortality risks, and Long-Term Care and Medical Expenditures</i>		
Ψ	Transition matrix of LTC-status	Mikoshiha et al. (2023)
$n^p(j^p, h)$	# of household members	Mikoshiha et al. (2023)
$H_{pm}(j^p)$	Avg. gross LTC expenditure	SLBE (2015) and Population Statistics
$M_i(j^p)$	Avg. gross medical expenditure	NME (2015)
<i>Endowments</i>		
Ω	Skill inheritance transition	Lefranc et al. (2014) and ESS (2017)
$\epsilon(j^k, z)$	Avg. earnings of married females	ESS (2017)
$\overline{WH}_{yf}(j^k)$	Avg. working hour	Time Use Survey (2016)
$\overline{DH}_{yf}(j^k)$	Avg. disposal hour	Time Use Survey (2016)
Θ	Persistence of wage shock	0.98 by Hsu and Yamada (2019)
σ_ζ	Variance of wage shock	0.09 by Hsu and Yamada (2019)
$y_{km}(j^k, z)$	Avg. earnings of married males	ESS (2017)
$\bar{l}_{km}, \bar{l}_{pm}, \bar{l}_{pf}$	Avg. leisure time	{0.54, 0.54, 0.50} by Time Use Survey (2016)
<i>Care Arrangement</i>		
$\bar{q}$	Avg. formal care cost in facility	327.83 (10,000-yen) by SIEL (2016)
$\bar{c}$	Avg. livings costs in facility	121.20 (10,000-yen) by SIEL (2016)
p	Avg. cost of FHC per hour	0.176 (10,000-yen) by CSLC (2016)
χ_h	Min. requirement of care hours	{1424.83, 3094.82} by CSLC (2016)
<i>Preference</i>		
σ	Risk aversion parameter	3.0
<i>Government</i>		
λ^h	LTCI copay	10 %
λ_{jp}^m	Public health insurance copay	30, 20, 10 % (varies by age)
κ	Pension replacement rate	1/3 by OECD (2019)
$\underline{c}$	Consumption floor	87 (10,000-yen) for widowed 123 (10,000-yen) for married
τ^c	Consumption tax rate	8 %
τ^l	Labor income tax rate	30 % by Gunji and Miyazaki (2011)
τ^a	Capital income tax rate	35 % by Kitao and Mikoshiha (2020)
<i>Other Parameters</i>		
r	Interest rate	2 % by Aoki et al. (2016)

Table 6: Jointly Estimated Parameters

Parameter	Value	Moment	Data	Model
<i>Preferences</i>				
β	0.98	Avg. per adult equivalent wealth	823.93	872.57
γ	0.48	Avg. female labor force participation rate	0.70	0.74
<i>Care Arrangements</i>				
A	1.85	Avg. annual care hours	1904.00	2334.18
ρ	0.66	IC-FHC hours correlation	-0.32	-0.48
$\theta(h = 2)$	0.36	IC hours ratio in total (Light LTC-status)	0.60	0.83
$\theta(h = 3)$	0.80	IC hours ratio in total (Heavy LTC-status)	0.48	0.45
ω	3.23	Ratio of those who use only IC	0.18	0.20
ξ	3.35	Ratio of recipients at home	0.61	0.57

6 Quantitative Analysis

In this section, I present the numerical results of the quantitative model. First, I review and discuss the results of the baseline model. I then analyze the roles of LTCI. This study evaluates how the universal LTCI with benefits-in-kind policy affects individual behavior and welfare by simulating policy experiments.

6.1 Baseline Model

In this section, I show the care arrangements at home in the baseline model. Table 7 shows the distribution of care arrangements at-home by LTC-status in the data and the baseline model. The calibrated model provides a good fit to the overall pattern of care arrangements. The mixed use of IC and FHC is the most common arrangement at home for each LTC-status. The use of FHC services increases with the level of care required, and the total annual hours of care vary significantly according to LTC status

Table 8 shows the distribution of care arrangements by family types in the baseline model. Family types depend on the skill combinations of the older parent generation z and the adult child generation z' in the baseline. For the ratio of IC hours, higher ratios are found for the low-skilled adult child generation and the high-skilled older parent generation. This is because the skill affects two key mechanisms that determine care arrangements: the opportunity cost of care and the amount of family savings.

In this model, the level of opportunity cost is determined by the permanent labor productivity $\epsilon(j^k, z')$ and idiosyncratic wage shocks μ . Since high-skilled individuals have higher permanent labor productivity, low-skilled individuals have lower opportunity costs. If the opportunity cost is relatively low, the IC cost would become lower and the family

Table 7: Distribution of Care Arrangements: Data and Baseline Model

		Data	Model
<i>Aggregate</i>	Distribution		
	Only IC	18.06%	19.71%
	Mix IC-FHC	75.42%	74.64%
	Only FHC	6.52%	5.65%
	Total Care Hours	1904.00	2334.18
<i>Light (SL1-CL2)</i>	Distribution		
	Only IC	20.87%	25.89%
	Mix IC-FHC	74.49%	74.11%
	Only FHC	4.64%	0.00%
	Total Care Hours	1424.83	1965.83
<i>Heavy (CL3-CL5)</i>	Distribution		
	Only IC	10.10%	0.00%
	Mix IC-FHC	78.05%	76.32%
	Only FHC	11.85%	23.68%
	Total Care Hours	3094.82	3508.97

would tend to use more IC services.

It is note that the family savings are important for care arrangements because savings provide a source of insurance against disability risks and bequests for the future descendants. Once female older parents face disability shocks, their family uses their savings to cover the substantial expenses for formal care services. Furthermore, the altruism of older parents also affects the decision. The desire to leave a bequest increases the family savings ([Lockwood \(2018\)](#)). Particularly, in the two-sided altruism model, the older parent generation can increase the future resources of their descendants by leaving a bequest, and the adult child generation also can prevent the cutting-off of their bequest of the older parent generation by providing IC, as discussed in [Groneck \(2017\)](#).

The amount of family savings depends on the skill level of the older parent generation, as in the last row. This is because adult children generation joins a family line with zero assets and the labor earnings of high-skilled individuals are much higher for both sexes, as shown in Figures 3 and 4. Then, the cost of choosing IC services becomes lower for the family with the high-skilled older parent generation. The higher amount of family savings leads to a higher level of consumption that the family would enjoy, and the family increases their overall utility by leaving a larger bequest for the descendants. In other words, the family with sufficient savings chooses to decline the current labor earnings through IC rather than smaller bequest for a savings cut-off by FHC services. In

Table 8: Distribution of Care Arrangements by Family Types in the Baseline Model

	Family Type (Skill Combinations)				Average
	HL	HH	LL	LH	
IC hours ratio	80.89%	74.60%	73.56%	66.30%	73.73%
Distribution					
Only IC	29.40%	49.52%	3.65%	21.11%	19.71%
Mix IC-FHC	66.93%	42.80%	92.99%	67.03%	74.64%
Only FHC	3.67%	7.68%	3.36%	11.87%	5.65%
Avg. per adult equivalent wealth	1503.09	1873.44	439.26	712.57	872.57

contrast, when family savings are not sufficient to purchase formal care services, families turn to IC or a welfare transfer program.

The first row shows that the family savings mechanism dominates the opportunity cost mechanism. This is due to the sufficient exogenous labor earnings of the married male in the adult child generation. As the male-female earnings gap is well documented in [Kitao and Mikoshiba \(2020\)](#), married men have extremely high labor force participation and earn the most.

Before moving on to the policy experiments, it is note that this model does not capture externally low-income families. Although the proportion of welfare transfer program recipients accounts for approximately 2% of the population in the real economy, the proportion at baseline accounts for only 0.36% of the population. This is because this model focuses on the family of two married couples. However, this seems reasonable to capture the characteristics of the family born between the 1950s and the 1980s.

6.2 Policy Experiments

I now study counterfactual policy experiments to evaluate the welfare effects of the universal LTCI with benefits-in-kind policy. I simulate the model under different scenarios from the baseline and compare the results. The welfare measure is calculated as the percentage change in consumption required in all possible states, ensuring that people are indifferent between the baseline and simulated scenarios.

In each scenario, this study first does not adjust lump-sum taxes. I set G in the first simulation to absorb the imbalance and satisfy the equation (4). This allows it to isolate the effects of tax adjustments and focus on changes in behavior of the heterogenous family. After that, this study adjusts a lump-sum tax rate to account for a change in net government revenue to balance the budget constraint.³⁰

³⁰Policy experiments in this chapter parallel that of [Fukai et al. \(2021\)](#) in that they first perform simulations without tax adjustment and with tax adjustment, and compares the two.

Table 9: Welfare Effects of Cash Benefits

		No Tax change	Tax adjusted
<i>Average Change</i>	IC users ratio	5.65 ppt	5.65 ppt
	IC hrs ratio	14.77 ppt	14.78 ppt
	At-home ratio	29.81 ppt	29.74 ppt
	FLFP	-3.49 ppt	-3.37 ppt
	Savings	-7.09%	-7.03%
	Welfare recipients	-0.17 ppt	-0.17 ppt
	Lump-sum tax (JPY)		3008.24
<i>Welfare Effects</i>	Average	0.75%	0.63%
	By Family Type		
	LL = (Low, Low)	0.86%	0.73%
	LH = (Low, High)	0.79%	0.69%
	HL = (High, Low)	0.51%	0.41%
	HH = (High, High)	0.46%	0.38%

Note: The table presents the changes in variables relative to those in a baseline model.

6.2.1 In-Kind Transfers versus Cash Benefits

To understand the role of LTCI with a benefit-in-kind policy, I first simulate an alternative scenario in which LTCI provides only cash benefits under 100 % copayment rate. I set the total government expenditure for cash benefits equal to the government expenditure on LTCI in the baseline model. In particular, the cash benefit amount is set to be equal to the total government spending in each LTC-status of the baseline, corresponding to 63.02 and 282.59 (10,000-yen) for light and heavy LTC-status, respectively.³¹ Table 9 presents the changes in the aggregate variables of the families' behavior and the welfare effects of heterogeneous families.

The cash benefit policy increases IC in terms of both the extensive and intensive margins due to the higher price of formal care services. Nursing home care services also become comparatively more expensive, leading more families to choose at-home care. By increasing the use of IC, the labor force participation of married females decreases. As cash transfers may compensate caregivers for the reduction in labor income, they will reduce incentives to save. With cash transfer compensation, the proportion of means-tested welfare recipients would be lower than in the baseline model. As a result, the welfare effects of no tax adjustment would be positive for all combinations of skill types. In particular, the welfare gains are higher for families with low productivity in both

³¹The same calculations are made for the cash benefit amount for male older parent in the no-disability status.

Table 10: Welfare Effects of Cash Benefits under LTCI Expenditure Cuts

	10 % LTCI spending reduction		20 % LTCI spending reduction	
	No Tax change	Tax adjusted	No Tax change	Tax adjusted
<i>Average Change</i>				
IC users ratio	5.65 ppt	5.65 ppt	5.65 ppt	5.65 ppt
IC hrs ratio	14.90 ppt	14.89 ppt	15.00 ppt	15.00 ppt
At-home ratio	27.44 ppt	27.36 ppt	25.00 ppt	24.88 ppt
FLFP	-3.29 ppt	-3.00 ppt	-2.87 ppt	-2.81 ppt
Savings	-4.06%	-3.98%	-0.86%	-0.77%
Welfare recipients	-0.05 ppt	-0.05 ppt	0.06 ppt	0.07 ppt
Lump-sum tax (JPY)		3100.00		3363.02
<i>Welfare Effects</i>				
Average	0.32%	0.19%	-0.11%	-0.25%
By Family Type				
LL = (Low, Low)	0.37%	0.23%	-0.11%	-0.27%
LH = (Low, High)	0.39%	0.28%	-0.11%	-0.13%
HL = (High, Low)	0.16%	0.05%	-0.17%	-0.30%
HH = (High, High)	0.17%	0.08%	-0.11%	-0.22%

Note: The table presents the changes in variables relative to those in a baseline model.

generations.

With tax adjustment, the second row of Table 9 shows that LTCI with cash benefits would require about 3008.24 yen from each individual. The increase of IC and decline in savings reduce the tax revenue from labor income of working-age married females and from capital income, thereby precipitating the imposition of a lump-sum tax. However, the level of the lump-sum tax is extremely modest since the productivity of married females is significantly lower than that of married males. This results in the positive welfare effects being slightly mitigated, but holds even after the imposition of a lump-sum tax. This experiment indicates that the cash benefit policy is more effective than the benefit-in-kind policy in terms of improving overall welfare. Furthermore, the limited impact of the reduction in tax revenues suggests that the positive welfare effects may be sustained even if government expenditures for the LTCI are reduced.

This study then considers policy experiments in which government cuts spending on LTCI by 10% and 20%, comparing the results with the baseline model. Table 10 reveals the welfare effects of cash benefits under the LTCI expenditure reduction: the amount of cash benefits decreases by 10% and 20% in each LTC-status, respectively. Compared to Table 9, the cash benefit reduction increases the share of informal care in total care hours,

while leaving the share of those choosing informal care unchanged. In addition, the effect of the reduction in labor force participation and savings is mitigated by the reduction in government benefits. The welfare effect remains positive even when cash transfers are reduced by 10%. Compared to before the reduction (Table 9), there is a slight increase in the number of welfare recipients, but the share of welfare recipients is lower than in the baseline. The second column also shows that the tax adjustment effect remains small. On the other hand, the welfare effect decreases as cash transfers decrease, and the welfare effect becomes negative for a 20% reduction.

These experiments suggest that it would be possible to maintain welfare effects while reducing government spending by providing cash transfers, given the limited impact of the reduction in tax revenues from cash benefits. It should be noted, however, that these values may represent upper bounds. This is because in countries that have introduced cash transfers, such as Germany, applicants for cash transfers are subject to screening and monitoring costs. In Germany, informal caregivers are officially qualified by the MDK (Medizinischer Dienst der Krankenversicherung) to receive cash transfers, in order to ensure the quality of care.

6.2.2 Universal Insurance versus Mean-Tested Welfare Program

This study also considers the role of the universal LTCI compared to the means-tested welfare program. This study simulates an extreme scenario removing LTCI from the baseline model. Table 11 shows the changes in the aggregate variables and the welfare effects of heterogeneous families. In the no tax adjustment scenario, both the intensive and extensive margins of IC and at-home use increase due to the higher cost of formal care. The additional increase in IC would reduce the labor force participation of working-age married females. At the same time, average savings would increase as the family would likely accumulate more precautionary savings. However, the higher expenditures would hit the family with lower savings, resulting in a higher share of welfare recipients. Thus, the welfare effects would be strictly negative in the no-tax-adjustment scenario. The welfare loss is substantial for families with both low-skilled children and parents.

Adjusting the lump-sum transfer to balance the budget, each family member would receive nearly 80,000 yen annually, leading to positive welfare effects. The compensation from the lump-sum transfer would increase IC because families would have an incentive to avoid saving an additional bequest. As a result, female labor force participation would be lower and average saving would be higher than in the no tax adjustment scenario. The increase in the number of means-tested welfare recipients is higher relative to the baseline, but lower than in the no tax adjustment scenario.

It should be noted, however, that this result depends on the generosity of the welfare program, as measured by the level of the consumption floor. Table 12 shows the welfare effects of eliminating LTCI for the baseline and for different consumption floor increases.

Table 11: Welfare Effects of Removing LTCI from Baseline

		No Tax change	Tax adjusted
<i>Average Change</i>	IC users ratio	5.65 ppt	5.65 ppt
	IC hrs ratio	12.04 ppt	12.74 ppt
	At-home ratio	8.68 ppt	12.69 ppt
	FLFP	-1.21 ppt	-3.96 ppt
	Savings	14.40%	16.23%
	Welfare recipients	2.29 ppt	1.86 ppt
	Lump-sum tax (JPY)		-80,505.77
<i>Welfare Effects</i>	Average	-3.71%	0.96%
	By Family Type		
	LL = (Low, Low)	-4.10%	0.95%
	LH = (Low, High)	-3.80%	0.37%
	HL = (High, Low)	-2.97%	1.05%
	HH = (High, High)	-2.64%	1.40%

Note: The table presents the changes in variables relative to those in a baseline model.

The means-tested welfare program has the effect of distorting the saving decision. As consumption levels rise, the tax-adjusted welfare effect declines. Eliminating LTCI results in reduced government spending, which is offset by increased spending on means-tested welfare programs. Consequently, the lump-sum transfer amount is inadequate to offset the significant care burden.

Table 12: Welfare Effects of Removing LTCI from Different Economies

	Baseline		Cons. Floor 20% up		Cons. Floor 50% up	
	No change	Adjusted	No change	Adjusted	No change	Adjusted
Average	-3.71%	0.96%	-3.78%	-0.21%	-3.33%	-0.96%
By Family Type						
LL = (Low, Low)	-4.10%	0.95%	-3.94%	-0.11%	-3.10%	-0.58%
LH = (Low, High)	-3.80%	0.37%	-4.02%	-0.83%	-3.63%	-1.54%
HL = (High, Low)	-2.97%	1.05%	-3.40%	-0.00%	-3.66%	-1.34%
HH = (High, High)	-2.64%	1.40%	-3.19%	-0.30%	-3.74%	-1.75%
Lump-sum tax (JPY)		-80,506		- 62,305		-42,282

Note: The table presents the changes in welfare effects of removing LTCI relative to those in economy with LTCI, respectively. The column labeled "No change" represents the change without the tax adjustment, and the column labeled "Adjusted" represents the change with the tax adjustment.

7 Conclusion

This study quantifies the welfare effects of LTCI on heterogeneous households relative to alternative policies and focuses on the role of universal LTCI with a benefits-in-kind policy. This study considers the Japanese economy, which has the highest aging rate in the world, and constructs a life-cycle model of family decision-making under long-term care risks. I develop a structural overlapping generations model with two-sided altruism. Based on empirical evidence on disability and mortality risks and care arrangements, my model focuses on the LTC-status of females in the older parent generation and the care arrangements between the female older parent and the female adult child as her primary caregiver. To develop a richer model of care arrangements, I incorporate three types of care options: IC from her adult child, FHC, and public NHC. This model endogenizes care arrangements by introducing two-stage family decisions.

This study focuses on the two features of Japan's LTCI system: mandatory universal insurance and a benefits-in-kind policy. This study provides two important insights. First, the welfare effects of in-kind and cash transfers depend on the productivity of caregivers. The in-kind policy is more costly than the cash policy, although the former does not significantly discourage the labor supply of caregivers. If the productivity of caregivers is very low and the impact of labor income tax losses is small, cash transfers can reduce government spending on LTCI and still maintain positive welfare effects. Second, the welfare effects of eliminating the LTCI system depend on the generosity of the welfare program. Because of the substantial caregiving burden and the absence of universal LTCI, families turn to IC or the means-tested welfare program. However, the means-tested welfare program distorts the saving decision, and the reduction in government spending on LTCI is offset by increased spending on the welfare program.

There are key concerns about—and omissions from—my model, which can be investigated in future research. The first concern is the impact of demographic change. Although this study focuses on the care arrangements between widowed females and their adult female children of working age, this assumption of family structure is changing with declining fertility rates for unmarried and later marriages, a declining trend in which daughters-in-law are the primary caregivers. It is important to consider the impact of these changes on the welfare effects of LTCI when considering policy reforms and the fiscal sustainability of the insurance system. The second concern is NHC services in the long-term care market. This study does not consider private NHC, but it is important for understanding the impact of LTCI on wealthy families. Private NHC provides higher quality care and housing, and high-income households tend to use these services. To take NHC services seriously, I need to consider the heterogeneity of demand for different types of NHC services.

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A The CSLC Data

This section provides more details about our use of the Comprehensive Survey of Living Conditions (CSLC) data. The CSLC is a nationally representative survey conducted by the Ministry of Health, Labour and Welfare (MHLW) in Japan. The survey was first conducted in 1986. The CSLC serves as comprehensive data sources covering families and their members nationwide.³²

The CSLC consists of five questionnaires: household, long-term care, health, income, and savings questionnaires. The household and income questionnaires are conducted annually, however, long-term care, health, and saving questionnaires are conducted only once every three years in the large-scale survey year. For the purpose of this study, this study uses the household and long-term care questionnaires from the 2016 large-scale survey. The household questionnaire provides key information about families and their members, including details on family structure, age, sex, and care demand. Meanwhile, the long-term care questionnaire contains information on care demand, including certification for long-term care or support, the level of care, type and costs of care services.

The 2016 CSLC was conducted on approximately 710,000 individuals, randomly sampled from 5,410 districts, as recorded in the 2010 National Census. Additionally, the 2016 long-term care questionnaires complementarily cover around 8,000 LTCI-certified individuals from 2,446 of the 5,410 districts.

A.1 Care Sample

This section provides descriptive statistics of care sample. The care sample is constructed from the CSLC by matching household questionnaires with long-term care questionnaires. The sample includes those eligible for LTCI services, aged between 65 and 94, but is limited to those who provide information on the level of care and primary caregivers. Table 13 presents characteristics of the older adults with disabilities, receiving care at home.

³²The following individuals are not covered by the CSLC: single workers; migrant workers; long-term business travelers (generally more than 3 months); students on leave; residents of nursing homes; long-term hospitalized patients (those whose resident registration has been transferred to a hospital); foster children left in care; incarcerated persons; and other separated persons.

Table 13: Descriptive Statistics of “care sample”

Variable of older adults with disabilities		Mean	Std Dev
<i>Age and Sex (N=5,181)</i>	Age	83.06	6.68
	Female	0.65	0.48
<i>LTC-status (N=5,181)</i>	Light (SL1-CL2)	0.73	0.45
	Heavy (CL2-CL5)	0.27	0.45
<i>Marital Status(N=5,181)</i>	Widowed	0.51	0.50
	Married	0.44	0.50
	Divorce	0.03	0.17
	Single	0.02	0.15
<i>Children (N=5,017)</i>	Having at least one child	0.91	0.29
<i>Living arrangements with children (N=4,372)</i>	Resides together	0.63	0.48
	Resides within the same municipality	0.24	0.43

Note: The data source is from care sample of the 2016 CSLC by the MHLW. The care sample has 5,181 observations, however, note that number of observations with information on children is 5,017, of which 4,372 have information on residence with children. The numbers in the table are derived from the author’s calculation and may not correspond to the numbers published by the MHLW.

As can be seen from the table above, the average age is 83.06 years, with females constituting 65.01% of the sample. This is consistent with findings presented in section 3.2. The median level of care demanded is SL2. Within the sample, 72.79% have light LTC-status, while 27.71% have heavy LTC-status. Notably, the proportion of individuals with light LTC-status in the care sample is greater than that in the long-term care claims data (65.30%).³³ This could be explained by the CLSC’s focus on the non-institutionalized population, as primary public NHC are accessible solely to older adults with heavy LTC-status.

Table 13 illustrates widowed and married individuals comprise approximately 94.66% of the sample. When narrowing the focus to those providing information about children, 4,562 individuals (90.39%) have at least one child out of 5,017. Additionally, among the 4,372 individuals who have at least one child and provide information on living arrangements with their children, 3,787 (86.62%) have at least one child who lives together or in the same municipality.³⁴

³³The data are taken from the Report Survey on Situation of Long-term Care Insurance Service conducted by the MHLW in 2016. Website: <https://www.mhlw.go.jp/topics/kaigo/osirase/jigyo/16/index.html> (in Japanese) (Accessed September 13, 2024).

³⁴The trend of living arrangements with children persists irrespective of disability. In the 2016 CSLC, approximately 79.76% of older adults having children have at least one child who lives together or in the same municipality. Additionally, confirming the trend from 1992 to 2019, the average proportion of older adults with at least one child living together or in the same municipality is 82.64%, gradually declined from 86.89% in 1992 to 78.37% in 2019. Furthermore, the breakdown illustrates a reduction in the proportion cohabiting, decreasing from 66.65% in 1992 to 53.07% in 2019, while the proportion with children residing separately in the same municipality has increased from 20.24% in 1992 to 25.30% in

A.2 Family Sample

This section describes characteristics of primary caregivers in the family sample by marital status and sex.

As can be seen from Table 14, more than 80% of the widowed older adults receive care primarily from their children, including both their own children and children-in-law. The demographic characteristics of caregiving children exhibit similarities between widowed males and females. The average age of their children (primary caregivers) ranges from the mid to late 50s for both their own children and children-in-law. While slightly over half of their own children are female, more than 95% of children-in-law are females. The traditional role of daughters-in-law as primary caregivers is undergoing a decline, shifting toward male caregivers such as unmarried sons and male spouses, as reported by Tokunaga et al. (2015). As for employment, their own children are more likely to be regular workers, while their children-in-law tend to be contingent workers.

Table 14 shows that the primary caregivers for married older adults are their spouses. When comparing married males and females, the latter are more likely to be cared for by their own children and children-in-law. The characteristics of their spouses (primary caregivers) show their average age in the late 70s and very low labor participation rate (about 10%). This indicates that care is provided primarily by those who have retired from the labor market.

2019. Note that the numbers are derived from the author's calculation and may not correspond to the numbers published by the MHLW.

Table 14: Characteristics of primary caregivers by marital status and sex

	Primary Caregivers				
	IC (own child)	IC (child-in-law)	IC (spouse)	IC (other)	FHC
<i>Widowed Females (N=1,865)</i>					
Distr. of primary caregivers	63.11%	25.29%	0.00%	1.88%	9.71%
Age	58.77	58.46	-	33.01	-
Female	55.14%	98.73%	-	48.10%	-
LFP	59.19%	58.44%	-	59.52%	-
% Regular	43.94%	22.14%	-	87.64%	-
% Contingent	32.52%	43.28%	-	6.13%	-
% Self Employment	16.40%	5.14%	-	6.24%	-
% Other Employment	7.15%	29.44%	-	0.00%	-
<i>Married Males (N=915)</i>					
Distr. of primary caregivers	21.20%	4.94%	68.09%	0.73%	5.05%
Age	53.77	55.88	76.07	31.28	-
Female	57.59%	98.39%	100.00%	56.31%	-
LFP	72.01%	65.55%	11.86%	36.77%	-
% Regular	39.56%	31.83%	19.12%	100.00%	-
% Contingent	33.30%	49.85%	19.57%	0.00%	-
% Self Employment	12.48%	2.31%	32.34%	0.00%	-
% Other Employment	14.66%	16.02%	28.87%	0.00%	-
<i>Married Females (N=606)</i>					
Distr. of primary caregivers	29.52%	10.42%	52.70%	0.46%	6.90%
Age	54.68	55.64	79.32	76.00	-
Female	60.40%	94.30%	0.00%	100.00%	-
LFP	66.04%	59.71%	15.83%	0.00%	-
% Regular	31.22%	36.75%	23.30%	-	-
% Contingent	47.47%	37.47%	13.42%	-	-
% Self Employment	16.20%	6.61%	56.25%	-	-
% Other Employment	5.11%	19.17%	7.04%	-	-
<i>Widowed Males (N=292)</i>					
Distr. of primary caregivers	59.78%	21.61%	0.00%	2.08%	16.53%
Age	55.36	54.00	-	26.00	-
Female	52.48%	96.99%	-	84.28%	-
LFP	71.18%	50.55%	-	82.48%	-
% Regular	50.19%	38.77%	-	100.00%	-
% Contingent	24.67%	40.18%	-	0.00%	-
% Self Employment	22.83%	1.70%	-	0.00%	-
% Other Employment	2.31%	19.35%	-	0.00%	-

Note: The data source is from care sample of the 2016 CSLC by the MHLW. The CSLC sample is constructed by matching household questionnaires with long-term care questionnaires, including those who are eligible for LTCI (65 to 94 years old) and those who are widowed or married with at least one child living together or in the same city. The sample is limited to observations with information on the level of care, primary caregivers, other caregivers and the use of FHC services. The figures are derived from the author's calculation and may not correspond to the numbers published by the MHLW.

B Two-sided Altruism and Strategic Motive

Previous empirical studies show a positive correlation between parental transfers (e.g. bequests and inter-vivos transfers) and child-provided IC. For example, in Japan, children who provide IC to their parents are more likely to receive a larger share of bequests than other children ([Horioka 2002](#)), and children who expect to receive bequests from their parents tend to live with them and have more contact with them ([Yamada 2006](#)). Various theoretical models have been discussed to explain these empirical facts. I discuss two main theoretical models: (i) two-sided altruism and (ii) strategic bequest motive.

In two-sided altruism, parents and children exhibit altruism toward each other ([Laitner 1997](#); [Laferrère and Wolff 2006](#)). The parental transfers of resources to children become altruistic behaviors to compensate for the loss of utility from the burden of child-provided IC. Additionally, altruistic children provide IC voluntarily. Models with two-sided altruism have dynastic structures: children inherit their family line and resources from their parents. In contrast, for strategic bequest motive, parents provide transfers to receive IC from children ([Bernheim et al. 1985](#); [Cox 1987](#)). When children do not voluntarily care for their parents as much as they would like, parents can transfer their resources to their children contingent on IC.

There are several studies on strategic bequest motive and child-provided IC. For example, comparing the patterns of bequest distribution when the first parent dies (primary inheritance) and the second parent dies later (secondary inheritance), [Hamaaki et al. \(2019\)](#) show that bequest motive and children's help in long-term care are consistent with dynastic motive. In particular, the empirical evidence from secondary inheritance suggests the existence of strong traditional family values. Note that Japanese parents divide their bequests unequally among their children and do not leave inter-vivos transfers or written wills.

However, as discussed in [Groneck \(2017\)](#) and [Nakamura and Maruyama \(2012\)](#), it is difficult to identify the motive for the substantial impact of caregiving on bequests. This is because both theoretical models can interpret the positive correlation between parental transfers and child-provided IC. Furthermore, unfortunately, as discussed in [Nakamura and Maruyama \(2012\)](#), many previous studies in Japan have concluded support for strategic bequest motive based solely on the positive correlation between parental transfers and children who provide IC and/or live with their parents.

I then discuss the theoretical limitations of two-sided altruism and strategic bequest motive. The two-sided altruism model is tractable compared to strategic motive. However, as discussed in [Mommaerts \(2025\)](#) and [Boar \(2021\)](#), two-sided altruism makes it difficult to analyze the timing of parental inter-vivos transfers and the different wealth accumulation paths of parents and children. In contrast, the estimation of strategic bequest motive requires detailed data on the existence of inter-vivos transfers and the savings

paths of children and parents. Both approaches have been used in previous structural models of long-term care. As parents in the U.S. tend to divide their bequests equally and leave inter-vivos transfers and written wills for unequal transfers, Mommaerts (2025) considers a cooperative framework with limited commitments and Ko (2022) considers non-cooperative decision-making between parents and children. Moreover, Barczyk and Kredler (2017) and Barczyk et al. (2022) consider both two-sided altruism and the bargaining process between parents and children by developing a dynamic non-cooperative framework. In contrast, Imrohoroglu and Zhao (2018) use two-sided altruism by focusing on within-family saving behavior using data in China.

As a result, this study chooses two-sided altruism because empirical evidence suggests that bequest motive and child-provided IC are consistent with two-sided altruism. Strong traditional family norms still exist in Japan. Additionally, inter-vivos transfers and/or written wills, which are important elements in estimating strategic bequests, are not widely observed in Japan. However, more research on the motive for the substantial impact of caregiving on received bequests is desirable because identifying the motive is important to analyze the saving behavior of elderly parents.

C Computation Algorithm of Steady State

In this section, I present the algorithms used to compute the steady state following the five steps described below.

Step 1: Guess $\text{pen}_{pf}(j^p, z)$ and τ^{ls} .

Step 2: Given the interest rate r and a set of government policies $\{\lambda^h, \lambda^m, \tau^c, \tau^a, \tau^l\}$, calculate the problem of the family. Guess then the value function of no-disability of age $j^k = 1$, $V_{j^k=1}^{ND}$, and solve the family problem by backward induction. Update the guess of $V_{j^k=1}^{ND}$ and iterate until convergence.

Step 3: Compute the set of age-dependent measures of family $\{X\}_{j^k=1}^J$ from the policy function in Step 2. Guess the age-dependent measures of age $j^k = 1$, X_1 , and calculate the age-dependent measures to satisfy equations (5) and (6). Update the guess of X_1 and iterate until convergence.

Step 4: Use the policy function and set of age-dependent measures of family and calculate aggregate variables.

Step 5: Use equation (3) and government budget conditions to update the guesses of $\text{pen}_{pf}(j^p, z)$ and τ^{ls} , if needed.